

Harvesting Factor Premia Across Regimes: An Anchor-Stabilized Hidden Markov Framework for Multifactor Portfolios

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Abstract

Two influential industry studies provide complementary benchmarks for dynamic factor allocation. Ilmanen et al. (2021) of AQR show that traditional factor-timing strategies often deliver limited out-of-sample gains, while Botte and Bao (2021) of Two Sigma show that latent market states can be recovered from factor-return panels but do not directly translate regime information into portfolio decisions.

This paper develops a regime-conditioned downside-aware multifactor allocation framework that converts latent market states into implementable portfolio weights. Focusing on a U.S. equity investment universe, we extend static Gaussian Mixture Model regime classification into a Gaussian Hidden Markov Model that captures regime persistence and transition dynamics. One-step-ahead regime probabilities are mapped into factor allocations through turnover-aware mean–variance optimization with Ledoit–Wolf covariance shrinkage. VIX and CAPE are excluded from regime identification and used only as auxiliary optimization anchors, stabilizing the reference allocation without serving as standalone return predictors.

Evaluated under a strict expanding-window walk-forward protocol from 2013–2023, the framework improves downside control and produces smoother return paths, especially during the 2020 COVID shock and the 2022 joint equity–bond drawdown. The S&P 500 achieves a higher annualized return over the same period, but with 15.06% volatility, a Sharpe ratio of 0.90, and a maximum drawdown of -23.97%. In comparison, a long-only protective specification achieves a Sharpe ratio of 1.33 with maximum drawdown of -3.64%, and a long-short specification achieves a Sharpe ratio of 1.24 with maximum drawdown of -5.78%. The results suggest that factor premia remain economically useful when latent regime information is translated into disciplined portfolio decisions.

Keywords: Factor Premia, Regime Switching, Hidden Markov Models, Dynamic Asset Allocation, Multifactor Investing, Portfolio Optimization.

1 Introduction

1.1 Motivation

This paper asks whether a transparent, rules-based smart-beta equity strategy can remain robust as market regimes switch more frequently than they did in earlier decades. Factor investing is motivated by the idea that characteristics such as value, momentum, quality, size, low risk, and profitability earn long-run premia. However, these premia are not earned smoothly. Individual factors can underperform for extended periods, experience sharp reversals, or become highly correlated during market stress. A static multifactor portfolio may therefore appear diversified

in normal periods but fail to provide adequate protection when factor leadership changes or when market regimes shift.

This concern has become especially relevant in the post-2008 environment. Financial markets have alternated between crisis episodes, central-bank-supported recoveries, inflation shocks, and rapid factor rotations. During such periods, the same factor exposure can have very different risk-return implications depending on the underlying state of the market. Momentum, for example, can perform well in persistent trends but suffer during sharp rebounds; value and quality may behave differently across recessions, recoveries, and high-rate environments; and low-risk or defensive factors may protect capital in drawdowns but underparticipate in sudden rallies. These patterns motivate a conditional approach to factor allocation.

A natural response is factor timing, but the empirical evidence on timing signals is mixed. Ilmanen et al. (2021) show that traditional factor-timing strategies often deliver limited out-of-sample gains after accounting for estimation uncertainty and realistic implementation. This does not imply that factor premia are no longer useful. Rather, it suggests that directly forecasting factor returns from macro variables is difficult and potentially fragile. A more robust approach may be to use regime information as an intermediate state variable: instead of asking whether a single macro signal predicts next month’s factor return, we ask whether the prevailing market environment changes the appropriate allocation across factors.

Recent regime-based work supports this direction while also motivating further development. Botte and Bao (2021) show that economically interpretable latent states can be recovered from factor-return panels using Gaussian Mixture Models. Their framework is valuable because it demonstrates that factor returns contain information about market regimes without requiring ex-ante regime labels. At the same time, the GMM treats observations as independent across time and is primarily used for regime characterization rather than real-time allocation. In deployment, this leaves two natural extensions: explicitly modeling regime persistence and developing a direct mechanism for translating regime information into portfolio weights.

A first related approach is the HMM-based factor-switching model of Wang, Lin, and Mikhelson (2020). Their framework provides a useful demonstration of how hidden Markov models can connect market-regime detection with factor-investing decisions. Specifically, they use broad market return and volatility variables to infer market regimes, and then rotate among pre-specified factor models according to their historical performance in each regime. This design is intuitive and operationally transparent, making it a helpful benchmark for studying HMM-based factor allocation. The present paper builds on this idea from a different angle. Rather than using regimes to select one model from a fixed menu, we use HMM-implied predictive regime probabilities as inputs to a continuous portfolio optimization problem. This allows the final allocation to reflect uncertainty across possible future regimes instead of relying on a single detected state.

A second closely related contribution is the dynamic factor allocation framework of Shu and Mulvey (2024). Their work is particularly relevant because it directly studies factor allocation through regime-switching signals. By applying sparse jump models to factor active returns, they obtain stable and interpretable factor-specific bull and bear regimes, and then translate these regime views into a Black–Litterman allocation across long-only factor indices. This provides an elegant and practically motivated way to connect factor cyclicity with portfolio construction. Our paper is complementary to their approach. While their framework emphasizes factor-specific two-state regimes and Black–Litterman view synthesis, we focus on shared latent regimes inferred from the joint cross-section of factor premia. This allows the model to capture broader market-state dependence across factors, permit more than two regimes, and translate predictive regime probabilities directly into regime-conditioned mean–variance allocations.

This paper therefore extends the existing literature by developing a probability-weighted allocation pipeline for regime-aware factor investing. The goal is not to replace prior regime-switching approaches, but to study a complementary implementation channel. The HMM

produces filtered posterior regime probabilities, which are projected one step ahead through the estimated transition matrix and used to form forward-looking allocations. These probabilities are not used only to identify the current regime or to select a single strategy. Instead, they weight regime-conditional mean–variance solutions, producing a continuous allocation that incorporates uncertainty over future market states. In this sense, the framework connects latent regime inference with portfolio construction through an optimization-based allocation layer.

The portfolio construction step is equally important. Regime detection alone does not create an implementable strategy. The regime signal must be translated into stable factor weights under estimation error, turnover constraints, and realistic rebalancing. We therefore combine regime-conditional mean–variance optimization with Ledoit–Wolf covariance shrinkage and an ℓ_1 turnover penalty. VIX and CAPE are used cautiously: they are excluded from the HMM likelihood so that regime identification is driven only by the factor–return panel, and they enter only as optimization anchors that stabilize the reference allocation. This design separates latent regime inference from macro timing while still allowing market-stress and valuation information to improve portfolio stability.

The resulting framework is best understood as a solving and testing pipeline for regime-aware factor allocation. It starts from raw factor returns, infers persistent latent regimes, projects regime probabilities forward, solves regime-conditional portfolio weights, stabilizes those weights with turnover and anchor controls, and evaluates the full process under a strict expanding-window walk-forward protocol. The contribution is therefore not simply identifying regimes, nor simply showing that factors are cyclical. The contribution is an implementable method for converting latent regime information into disciplined multifactor portfolio decisions.

1.2 Contributions

We position our contributions primarily relative to the GMM-based regime characterization of Botte and Bao (2021) and the factor-timing evidence of Ilmanen et al. (2021). These two papers provide complementary benchmarks: the former shows that latent regimes can be recovered from factor–return panels, while the latter highlights the practical difficulty of converting conditional factor–premia signals into robust out-of-sample allocation gains. Additional regime-aware allocation papers are discussed in Section 2; here we summarize how our framework extends these two anchors along four dimensions.

1. **From i.i.d. mixture classification to Markov-persistent regimes.** Rather than relying on observable-signal timing methods of the type examined in Ilmanen et al. (2021), we adopt a latent-state approach inspired by the GMM regime characterization of Botte and Bao (2021). Their Gaussian Mixture Model demonstrates that economically interpretable regimes can be recovered from factor–return panels, but the model treats monthly observations as conditionally independent and therefore does not explicitly represent regime persistence or transition dynamics. We extend this framework by fitting a Gaussian Hidden Markov Model initialized from the GMM estimates. The HMM is applied to the U.S. equity factor–premia panel and selected using expanding-window validation based on log-likelihood, AIC, and BIC diagnostics. In this setting, regime persistence and transition probabilities become estimated components of the model rather than imposed assumptions or omitted dynamics.
2. **From regime characterization to implementable allocation.** Botte and Bao (2021) focus on regime identification and economic interpretation. We extend this characterization step into a portfolio-construction problem by solving a regime-conditional mean–variance program with an ℓ_1 turnover penalty. We compare sample, Ledoit–Wolf, and exponentially weighted covariance estimators, and obtain a regime-to-weights map $W_{\text{regime}} \in \mathbb{R}^{K \times N}$. The final portfolio is formed by weighting these regime-conditional allocations using the HMM’s predicted regime probabilities. This links latent regime information directly to tradable multifactor weights rather than stopping at regime labels.

3. **A regime-conditioned optimization anchor.** Following the caution in Ilmanen et al. (2021), we do not use VIX or CAPE as standalone timing signals to directly forecast future factor premia. Instead, these variables enter only inside the portfolio optimization step as stabilizing anchors. After the HMM estimates latent regime probabilities from the factor-return panel, VIX- or CAPE-based deviations adjust the optimizer’s reference allocation w_0 through clipped regime-conditioned anchor schemes. The soft anchor applies posterior-weighted adjustments across regimes, while the hard anchor uses the dominant inferred regime. This design keeps regime identification separate from macro timing, while still allowing market-stress or valuation information to improve the stability of portfolio weights. Empirically, the anchor improves risk control for the long-only specification and improves the return-seeking profile of the long-short specification.
4. **A strictly out-of-sample forward-looking evaluation framework.** Instead of allocating based on contemporaneous regime labels, we project the filtered posterior distribution one step ahead through the estimated transition matrix and allocate using

$$\mathbf{w}_{t+1} = (\mathbf{P}_t \mathbf{T}) \mathbf{W}_{\text{regime}},$$

so that portfolio weights are formed using only information available at time t . The full 1990–2023 sample is partitioned into three non-overlapping roles: an initial fitting window from 1990–1999 for anchor regression coefficients, a hyperparameter validation window from 2000–2012 for HMM specification and anchor configuration selection, and a held-out test window from 2013–2023. During the held-out period, the HMM is refit monthly using an expanding window, while portfolio weights are reoptimized bi-monthly under the frozen specification. This design mirrors real-time deployment: regime inference, transition projection, anchor regression, and portfolio optimization all use only information available at each decision point. Relative to factor-timing studies that may estimate some parameters using long samples before reporting out-of-sample portfolio results, our framework applies a leakage-controlled treatment throughout model selection, HMM estimation, and walk-forward allocation.

2 Related Work

This paper has two primary anchors in the literature: Botte and Bao (2021) for latent regime characterization and Ilmanen et al. (2021) for conditional factor premia evaluation. Both are discussed extensively in Section 1, so this section places the paper in the broader literatures on regime detection, factor cyclicalities, regime-aware factor allocation, and portfolio construction under estimation uncertainty.

Regime detection in financial markets. The canonical approach to regime modeling is the Markov-switching framework of Hamilton (1989), which introduced regime-dependent dynamics to macroeconomic time series and was subsequently extended to equity returns, volatility, and international asset allocation (Ang and Bekaert, 2002). A parallel literature treats regime identification as an unsupervised learning problem, applying Gaussian Mixture Models, Hidden Markov Models, or related clustering methods to financial return panels without imposing ex-ante labels. The closest industry precedent to our setting is Botte and Bao (2021), who recover four economically interpretable market conditions from a U.S. factor-return panel using a Gaussian Mixture Model. Their framework is valuable because it shows that latent market states can be inferred directly from the joint behavior of factor premia. Our paper builds on this insight by introducing an HMM transition structure, so that regime persistence and transition probabilities are explicitly estimated rather than treating each monthly observation as independent.

Factor investing and state-dependent premia. A substantial body of work documents that factor premia vary with market and macroeconomic conditions. Daniel and Moskowitz (2016) show that momentum can suffer severe crashes during market rebounds following bear states, while Asness et al. (2013) document value and momentum premia across asset classes and emphasize their time-varying behavior. Subsequent work on dynamic factor timing examines whether macro indicators, valuation spreads, volatility measures, or sentiment proxies can be used to tilt factor exposures conditionally. Ilmanen et al. (2021) provide an important cautionary benchmark, showing that although some timing variables contain information, the realized out-of-sample gains from factor timing are often limited once estimation error and implementation frictions are considered. Following this caution, our framework does not treat VIX or CAPE as standalone factor-return predictors. Instead, the latent regime inferred from the factor cross-section serves as the main conditioning state, while VIX and CAPE enter only as auxiliary anchors in the portfolio optimization step.

Regime-aware factor allocation. Recent work has begun to connect regime detection more directly to factor allocation. Wang, Lin, and Mikhelson (2020) provide a useful HMM-based example: they infer broad market regimes from market return and volatility variables, and then rotate among pre-specified factor models according to their historical performance in each regime. Their framework is intuitive and operationally transparent, and it demonstrates how HMM regime detection can be linked to factor-investing decisions. Shu and Mulvey (2024) develop a closely related dynamic factor allocation framework using sparse jump models. They identify factor-specific bull and bear regimes from active factor returns, and then translate these regime views into a Black–Litterman allocation across long-only factor indices. This provides an elegant and practically motivated way to connect factor cyclicalities with portfolio construction. Our paper is complementary to these approaches. Rather than using regimes to select one strategy from a fixed menu, or estimating separate two-state regimes for each factor, we infer shared latent regimes from the joint cross-section of factor premia and map one-step-ahead predictive regime probabilities directly into regime-conditioned mean–variance allocations.

Portfolio construction under uncertainty. The mean–variance formulation of Markowitz (1952) remains the workhorse of portfolio allocation, but optimized weights are sensitive to noisy expected-return and covariance estimates. Two extensions are especially relevant here. First, covariance shrinkage improves stability when the number of assets is large relative to the available sample; Ledoit and Wolf (2004) show that shrinking the sample covariance matrix toward a structured target can improve out-of-sample performance. Second, turnover penalization helps reduce excessive trading caused by small changes in estimated inputs. Our framework combines both tools inside a regime-aware allocation loop: regime-conditional portfolios are solved using mean–variance optimization with Ledoit–Wolf covariance shrinkage and an ℓ_1 turnover penalty. VIX and CAPE are used to adjust the optimizer’s reference allocation rather than to generate direct trading signals or to enter the HMM likelihood.

Overall, the closest existing studies provide three complementary foundations: GMM-based characterization of latent factor regimes, HMM-based switching among factor models, and jump-model-based factor allocation through Black–Litterman views. The contribution of this paper is to combine HMM-based regime persistence, one-step-ahead regime-probability projection, regime-conditional optimization, turnover control, and anchor-based stabilization into a single leakage-controlled walk-forward allocation pipeline. This positioning does not replace prior regime-switching approaches; rather, it studies a complementary implementation channel for converting latent regime information into disciplined multifactor portfolio weights.

3 Data

3.1 Factors

- **Source:** Global factor data library from Jensen et al. (2023).
- **Sample period:** January 1984¹ – December 2023.
- **Frequency:** Monthly frequency is adopted for three reasons:
 - **Regime stability:** Regimes are macroeconomic and cycle objects rather than intraday phenomena. Monthly frequency better matches the timescale at which regime transitions occur.
 - **Signal-to-noise reduction:** Daily factor returns are heavily influenced by transient order flow and microstructure noise. Monthly frequency is the standard horizon in the academic factor literature, including Fama–French, Carhart, AQR, and Jensen, Kelly, and Pedersen (JKP).
 - **Strategy implementation realism:** The project focuses on a smart beta regime-aware multifactor premia strategy rather than high-frequency trading. Monthly rebalancing keeps turnover and transaction costs at levels consistent with real-world smart-beta implementation, whereas daily rebalancing would be operationally aggressive and invalidate the assumed trading-cost structure.
- **Universe:** US equities, using NYSE 20th percentile breakpoints and the non-microcap universe.
- **Factor construction:** Each factor is formed as a long–short tercile portfolio sorted on a firm characteristic, with accounting variables lagged four months and sign conventions aligned with the literature. Within each tercile, stocks are weighted using the capped value-weighted methodology of Jensen et al., where market capitalizations are winsorized at the NYSE 80th percentile before portfolio weights are computed. The factor returns are excess returns in USD.
- **Selected Factors:** Fourteen factors: accruals, debt issuance, investment, low leverage, low risk, mkt, momentum, profit growth, profitability, quality, seasonality, short term reversal, size, and value, that span the JKP theme clusters, concatenated with the market-return series to form a balanced matrix $T \times K$ with $T = 480$, representing a time span of 480 months; $K = 14$, representing number of selected factors.
- **Missing data treatment:** Only months with contemporaneous availability across all factors are retained, yielding a balanced panel. The sample period contains no missing data.
- **Scaling:** Returns are used in raw form for portfolio construction and evaluation, and standardized to zero mean and unit variance for GMM/HMM regime clustering estimation.

3.2 VIX

- **Source:** CBOE Volatility Index (VIX), retrieved from Yahoo Finance (ticker ^VIX).
- **Sample period:** January 1990 – December 2023, dictated by VIX inception in 1990; pre-1990 portions of the factor sample rely on the no-anchor specification.

¹For the general HMM regime characterization in section 5.1 we use the full 1984–2023 sample; the portfolio HMM and portfolio construction in section 5.2 use the 1990–2023 subset, dictated by VIX inception.

- **Frequency:** Monthly basis. Resampled daily closing levels to month-end to align with the factor panel.
- **Role in the framework:** Used exclusively as an auxiliary market-stress signal in the VIX-anchored optimizer variants; not an input to regime fitting.
- **Treatment:** Deliberately excluded from the factor panel used for clustering, so that its scale (a level in the 10–80 range, against factor returns in the percent range) cannot dominate the GMM/HMM likelihood, consistent with findings in Ilmanen et al. (2021). Enters the pipeline only through the optimizer’s warm-start w_0 .

3.3 CAPE

- **Source:** Cyclically Adjusted Price-to-Earnings Ratio (CAPE), retrieved from Robert Shiller’s online data library.
- **Sample period:** January 1990 – December 2023, matched to the VIX sample period for consistency and direct comparability across anchor specifications.
- **Frequency:** Monthly basis. Month-end CAPE observations were aligned with the factor panel.
- **Role in the framework:** Used exclusively as an auxiliary valuation-sensitive conditioning variable in CAPE-anchored optimizer variants; not an input to regime fitting.
- **Treatment:** Deliberately excluded from the factor panel used for clustering so that long-horizon valuation levels do not distort the GMM/HMM likelihood or dominate latent-state identification. Enters the pipeline only through the optimizer’s warm-start w_0 as part of the conditioning-anchor mechanism.

3.4 Factor Universe and Summary Statistics

Following the factor-panel framework in Botte and Bao (2021), we construct a diversified U.S. equity factor universe rather than modeling individual stock returns directly. The goal is to study whether latent market regimes can be identified from the joint behavior of economically distinct factor premia. The 14 factors are selected to cover several major dimensions of the equity factor literature, including market exposure, value, size, momentum, risk-based factors, profitability and quality, investment-related factors, financing activity, short-term reversal, and seasonality. This broad factor set is useful because regime changes may affect different factor families in different ways. For example, market stress may influence low-risk and quality factors differently from value, size, or momentum. Therefore, the selected universe provides a compact but economically diverse panel for regime identification and portfolio allocation.

Table 1: Summary statistics for the U.S. equity factor panel, January 1990 – December 2023.

Factor	Cum. Ret (%)	Ann. Ret (%)	Ann. Vol (%)	Max DD (%)	Kurtosis	Skewness
accruals	127.91	2.13	3.77	-21.10	0.78	0.16
debt_issuance	142.65	2.25	2.61	-9.96	12.07	1.44
investment	209.56	3.15	8.14	-28.17	7.43	0.82
low_leverage	-36.00	-0.38	12.14	-70.59	15.78	0.68
low_risk	66.82	2.32	14.34	-63.04	7.67	-0.22
mkt	1609.19	8.68	17.48	-55.14	2.64	-0.75
momentum	235.20	3.74	11.88	-40.50	12.08	-0.50
profit_growth	63.94	1.31	3.85	-19.39	5.76	-0.97
profitability	326.93	3.97	8.27	-32.90	10.76	0.50
quality	334.54	3.83	5.56	-15.68	1.49	-0.11
seasonality	73.41	1.40	2.09	-6.99	2.49	0.33
short_term_reversal	46.62	1.06	4.50	-23.24	18.22	0.90
size	-19.86	-0.26	7.70	-45.36	3.19	0.84
value	299.38	4.29	12.75	-61.89	14.60	-0.17

Table 1 reports cumulative return, annualized return, annualized volatility, maximum draw-down, skewness, and excess kurtosis for each factor over the full sample. The panel exhibits substantial heterogeneity across factors. The market factor achieves the strongest cumulative return, while several long-short factors, including low_leverage and size, display negative full-sample performance. This dispersion is useful for regime modeling because the HMM is intended to learn from cross-factor variation rather than from a single dominant return source.

The distributional statistics also show that factor returns are far from Gaussian at the individual-factor level. Excess kurtosis is elevated across much of the panel, particularly for short_term_reversal, low_leverage, value, and debt_issuance, consistent with the heavy-tailed nature of factor returns. We do not view this as a limitation for Gaussian mixture or hidden Markov modeling. Instead, the mixture structure is useful precisely because heterogeneous components can absorb tail behavior through high-variance regimes, rather than treating extreme observations as isolated outliers. In this sense, the factor universe follows the motivation of Botte and Bao (2021): a diversified factor panel provides information about latent market states that may not be visible from aggregate market returns alone.

Table 2: VIX summary statistics by decade.

Decade	Mean	Median	Std. dev.	Max
1990s	18.55	17.09	6.16	44.28
2000s	21.90	20.77	9.13	59.89
2010s	17.09	15.90	5.46	42.96
2020s	23.30	21.02	8.07	53.54

Table 2 shows visible VIX level shifts across the four decades — elevated through the late 1990s and 2000s, suppressed during the 2010s, and re-elevated post-2020 — motivate the choice to standardize the anchor against an in-regime mean rather than a full-sample mean.

4 Methodology

4.1 Regime identification

4.1.1 GMM

We first fit a Gaussian mixture model (GMM) to identify latent market states through cross-sectional differences in factor-return distributions, with respect to the benchmark model Botte

and Bao (2021) and a preparation for HMM initialization. The model assumes $p(x_t) = \sum_{k=1}^K \pi_k \mathcal{N}(x_t; \mu_k, \Sigma_k)$, where $\pi_k = P(z_t = k)$ denotes the prior probability (mixture weight) of latent regime k , and (μ_k, Σ_k) characterize the corresponding regime-specific mean and covariance structure.

Regime assignment follows a Bayesian posterior updating framework. Conditional on the observed factor-return vector x_t , the posterior probability that observation t belongs to regime k is:

$$P(z_t = k \mid x_t) = \frac{\pi_k \mathcal{N}(x_t; \mu_k, \Sigma_k)}{p(x_t)} \quad (1)$$

This produces a soft probabilistic clustering framework rather than hard regime classification. The GMM provides a flexible distributional clustering framework capable of capturing heteroskedastic and non-Gaussian aggregate behavior through mixtures of Gaussian components.

4.1.2 HMM

We extend the static GMM framework to a Gaussian hidden Markov model (HMM) in order to incorporate temporal persistence in regime dynamics. Let $s_t \in \{1, \dots, K\}$ denote the latent regime at time t . The latent state is assumed to follow a first-order Markov process with transition matrix $\mathbf{T}$, where

$$T_{ij} = P(s_t = j \mid s_{t-1} = i), \quad \sum_{j=1}^K T_{ij} = 1, \quad (2)$$

$$T_{ij}^{(0)}(d) = \begin{cases} d, & i = j, \\ \frac{1-d}{K-1}, & i \neq j. \end{cases}$$

In the implementation, the initial transition matrix is parameterized by a diagonal-strength parameter d , which controls the initial degree of regime persistence: A larger value of d initializes the HMM with stronger persistence, since the initial probability of remaining in the same regime is higher. Conversely, a lower value of d corresponds to a less-persistent initialization, assigning more probability to transitions across regimes. This is the key dynamic component that distinguishes the HMM from the static GMM, where latent regime assignments are governed by time-invariant mixture weights rather than a transition process. In the expanding-window refitting procedure, $T^{(0)}$ serves only as the starting value; the transition matrix is subsequently re-estimated using the data available in each training window.

Conditional on the latent regime, the observed factor-return vector follows a regime-specific Gaussian emission distribution,

$$x_t \mid s_t = k \sim \mathcal{N}(x_t; \mu_k, \Sigma_k). \quad (3)$$

Thus, relative to the GMM, the HMM preserves the same regime-specific Gaussian structure but replaces independent mixture assignments with a dynamic transition process across regimes. We initialize the HMM using the fitted GMM parameters (π, μ, Σ) , while the transition matrix $\mathbf{T}$ is specified through the model-selection procedure described below. Regime inference follows a Bayesian filtering framework. Given information available up to time t , the filtered posterior probability of regime k is

$$\alpha_t(k) = P(s_t = k \mid x_{1:t}) \quad (4)$$

which can be updated recursively as

$$\alpha_t(k) = \frac{\mathcal{N}(x_t; \mu_k, \Sigma_k) \sum_{j=1}^K \alpha_{t-1}(j) T_{jk}}{\sum_{\ell=1}^K \mathcal{N}(x_t; \mu_\ell, \Sigma_\ell) \sum_{j=1}^K \alpha_{t-1}(j) T_{j\ell}} \quad (5)$$

This filtering step uses only information available through time t , avoiding the look-ahead bias associated with smoothed probabilities that condition on the full sample. For portfolio allocation, we project the filtered posterior one step ahead using the transition matrix,

$$P(s_{t+1} = k \mid x_{1:t}) = \sum_{j=1}^K \alpha_t(j) T_{jk} \quad (6)$$

4.1.3 Model selection

The out-of-sample evaluation is implemented as a walk-forward backtest with a 23-year burn-in. During the burn-in, a GMM is fit on the return series to obtain initial estimates of the component means, covariances, and mixture weights, which seed the HMM's emission parameters. After the burn-in, at each month t the HMM is refit on the expanding window through t , filtered regime probabilities $\gamma_t(j) = P(s_t = j \mid \mathcal{F}_t)$ are obtained via the forward algorithm, and the one-step-ahead regime distribution $P(s_{t+1} \mid \mathcal{F}_t) = \gamma_t^\top \mathbf{T}$ is used to construct the next period's portfolio weights.

Selection criterion We jointly inspect the average out-of-sample log-likelihood, AIC, and BIC over $K \in \{2, \dots, 7\}$ for both the GMM and the HMM. Since AIC and BIC continue to decline across the tested range without exhibiting a clear minimum, we rely on an elbow inspection across both criteria to trade off fit against parsimony. This yields $K = 4$, consistent with the regime count reported in Botte and Bao (2021).

Hyperparameters We tune two hyperparameters of the HMM's emission and transition structure:

- **Covariance matrix structure:** we consider both full and diagonal emission covariances. With 14 factors, each state's full covariance matrix has $14 \times 15/2 = 105$ free parameters, which exceeds the effective per-state sample size available in the burn-in window. We therefore restrict the emission covariances to be diagonal.
- **Transition-matrix initialization:** We grid-search $d \in \{0.20, 0.22, 0.23, 0.24, 0.25, 0.30, 0.40\}$, setting $T_{ii}^{(0)} = d$ and $T_{ij}^{(0)} = (1 - d)/(K - 1)$ for $i \neq j$. Thus, d controls the initial regime-persistence strength only; the transition matrix is re-estimated in each expanding-window HMM refit.

Final configuration Number of regimes $K = 4$, diagonal emission covariance, and regime persistence strength $d = 0.23$.

4.2 Regime-conditional portfolio optimization (Burn-in)

For each regime $r \in \{0, 1, 2, 3\}$, we restrict the sample to in-regime observations within the burn-in window (noted that we unify the burn-in window based on the data availability: since VIX is the most limited and only started at 1990, so we set the starting point of the burn-in to be 1990) and solve the following ℓ_1 -regularized mean-variance problem once per regime:

$$\begin{aligned}
\min_{w \in \mathbb{R}^N} \quad & -\mu_r^\top w + \lambda w^\top \Sigma_r w + \alpha \|w - w_0\|_1 \\
\text{s.t.} \quad & \mathbf{1}^\top w = 1, \\
& w_{\min} \leq w_j \leq w_{\max}, \quad \forall j.
\end{aligned} \tag{7}$$

The mean-variance form is adopted for its tractability and interpretability: the objective is convex, λ has a direct interpretation as risk penalty, and only the first two moments of returns are required — both reliably estimable from a monthly panel without imposing distributional assumptions on higher-order moments. It remains the canonical objective in smart-beta and factor-portfolio construction, providing comparability to the existing literature.

Here μ_r and Σ_r are the in-regime mean and covariance estimated on the full burn-in sample, and $w_0 = 0$ during burn-in so that the ℓ_1 term acts as a sparsity regularizer (in the walk-forward variant, w_0 is set to the prior-period weights, repurposing the same term as a turnover anchor). The optimization yields a frozen $4 \times N$ matrix of regime-conditional weights $\{w_r\}_{r=0}^3$, which are then applied across the burn-in via the filtered regime probabilities $w_t = \sum_r \gamma_t(r) w_r$ to produce the performance series used for hyperparameter selection.

From regime weights to portfolio returns. At each month t in the burn-in evaluation window, we form the portfolio weight as a probability-weighted mixture of the regime-conditional solutions, using the HMM’s one-step-ahead predictive distribution:

$$w_{t+1} = \sum_{r=0}^3 \hat{\pi}_{t+1|t}(r) w_r, \quad \hat{\pi}_{t+1|t} = \gamma_t^\top \mathbf{T}, \tag{8}$$

where γ_t denotes the filtered regime probabilities at month t and $\mathbf{T}$ is the HMM transition matrix. The realized portfolio return is then $r_{p,t+1} = w_{t+1}^\top r_{t+1}$, and the series $\{r_{p,t}\}$ produces the Sharpe ratio, annualized return, maximum drawdown, and turnover statistics that feed the composite score.

Burn-in evaluation versus deployment evaluation. The burn-in evaluation fits the regime model once on the full burn-in window and computes filtered regime probabilities γ_t for all burn-in months from this single fit. Conditional on these inferred regime probabilities, regime-specific portfolio weights are then estimated using expanding-sample updates. This design reflects a sample-size constraint rather than a computational shortcut. With 276 monthly observations in the burn-in window and $K = 4$ regimes, each regime contains approximately 69 observations on average, which is already close to the lower bound for stable estimation of regime-specific emission parameters under the diagonal covariance restriction. A strict expanding-window refit within the burn-in period, starting for example from an initial 60-month window, would leave only about 15 observations per regime in early refits. This would generate unstable regime labels that may flip across consecutive refits and contaminate the hyperparameter ranking with estimation noise rather than genuine configuration differences.

Therefore, we use in-sample regime inference within the burn-in period, while treating burn-in performance only as a ranking signal for hyperparameter configurations rather than as a deployment-realistic performance estimate. Under the strict walk-forward protocol described in Section 4.5, the regime model is refit monthly on expanding data, ensuring no leakage in the real strategy tests. By the start of the deployment period in 2013, each refit uses at least the 276 burn-in observations plus all accumulated post-burn-in data, providing a more stable basis for HMM estimation. Portfolio allocation then uses one-step-ahead projected probabilities, $\gamma_t \mathbf{T}$, rather than same-period in-sample probabilities. Burn-in Sharpe ratios are therefore expected to exceed walk-forward Sharpe ratios under the same configuration. The size of this gap is itself

informative, as it indicates how strongly a model depends on in-sample regime inference, which is examined in the HMM–GMM comparison in Section 6.3.

Hyperparameters. The optimization is implemented in `cvxpy` (for the conic solvers) with a `scipy.optimize` fallback (for SLSQP). We grid-search over:

- **Solvers:** OSQP, CLARABEL, ECOS, and SLSQP.
- **Covariance estimators:** sample covariance (with 10^{-6} jitter and symmetrization for conditioning), Ledoit–Wolf shrinkage, and exponentially weighted with span = 36.
- **Risk aversion:** $\lambda \in \{0.2, 0.3, 0.4, 0.5\}$.
- **Weight bounds:** $w_{\min} \in \{-0.2, -0.1, -0.05, 0\}$, minimum weight of a single factor (light shorting permitted in the walk-forward variant); $w_{\max} \in \{0.4, 0.5\}$, maximum weight of a single factor.

The turnover penalty α is fixed at 0.01. Preliminary sensitivity checks showed that the objective and resulting weight trajectories are essentially invariant to α in a neighborhood of this value, so we did not include it in the grid.

Selection criterion (for burn-in). For a more informative and comprehensive selection, we further rank the configurations by a composite score combining four standardized burn-in performance metrics on top of plain return:

$$\text{Score}(\text{aggressive version}) = 0.2 \tilde{s} + 0.4 \tilde{r} - 0.2 \tilde{d} - 0.2 \tilde{t}, \quad (9)$$

$$\text{Score}(\text{protective version}) = 0.4 \tilde{s} + 0.2 \tilde{r} - 0.2 \tilde{d} - 0.2 \tilde{t}, \quad (10)$$

where $\tilde{s}, \tilde{r}, \tilde{d}, \tilde{t}$ are the z -scores (computed across grid configurations) of the burn-in Sharpe ratio, annualized return, maximum drawdown magnitude, and turnover, respectively. The weights reflect the strategy’s objective — a risk-controlled smart-beta allocation — by placing primary emphasis on risk-adjusted performance, reinforcing it with realized return, and penalizing deep drawdowns; turnover enters with a smaller weight because the portfolio rebalances bi-monthly, so trading costs are a secondary concern. Pure Sharpe is insufficient as a sole criterion: the highest-Sharpe configuration in the grid achieves its ratio by contracting both return and volatility by roughly 20%, and would contribute substantially less to terminal wealth than the composite-score winner.

Anchor selection. The anchor configuration selection also happens in the burn-in. See Section 4.4 for the comprehensive anchor analysis.

4.3 Portfolio weight projection and update (Walk-forward)

After the burn-in session, the framework is deployed monthly through 2023 under a strict walk-forward protocol. At each month t , the HMM regime clustering model is refit on the expanding history through month $t - 1$ to obtain updated emission parameters $\{\mu_r, \Sigma_r\}_{r=0}^3$ and transition matrix $\mathbf{T}$, and filtered regime probabilities γ_t are computed via the forward algorithm using information available through t inclusive. Every two months, the regime-conditional weights $\{w_r\}_{r=0}^3$ are re-solved by reapplying the optimization in Eq. (7) to the current in-regime expanding samples, with w_0 now set to the VIX- or CAPE-adjusted reference defined in the Anchor Scheme of Section 4.4; the ℓ_1 penalty therefore pulls solutions toward a sentiment-aware reference rather than acting as the sparsity regularizer it was during burn-in. Note that all of these configurations, including the anchor, are decided in the burn-in period. The walk-forward stage purely tests strategy performance to avoid leakage.

Anticipatory allocation. The deployed weight for month $t + 1$ is formed by mixing the regime-conditional solutions through the HMM’s one-step-ahead first-order Markov process transition matrix $\mathbf{T}$, further preventing look-ahead bias:

$$w_{t+1} = \sum_{r=0}^3 \hat{\pi}_{t+1|t}(r) w_r, \quad \hat{\pi}_{t+1|t} = \gamma_t^\top \mathbf{T}. \quad (11)$$

The allocation is therefore anticipatory in the persistence sense — it averages over the regimes the HMM expects to inhabit at $t + 1$ given evidence through t , rather than reacting to whichever regime is dominant now. The realized portfolio return is $r_{p,t+1} = w_{t+1}^\top r_{t+1}$.

Dominance switch. An optional `dominate` flag collapses the soft mixture to a one-hot allocation when a single regime overwhelmingly dominates the predictive distribution, i.e. $\max_r \hat{\pi}_{t+1|t}(r) > 0.95$, in which case $w_{t+1} = w_{r^*}$ with $r^* = \arg \max_r \hat{\pi}_{t+1|t}(r)$. This safeguard is inactive under the soft-mixture default used in our main results.

4.4 Anchor Scheme

Following Ilmanen et al. (2021), we treat macro-conditioning variables cautiously. VIX and CAPE are considered because they capture distinct economic states: VIX reflects near-term market stress and risk aversion, while CAPE reflects longer-horizon valuation conditions. However, consistent with the caution in Ilmanen et al. (2021) that factor timing signals often have limited and statistically fragile out-of-sample predictability, we do not use VIX or CAPE as standalone factor-return forecasting signals. They are also not included as inputs in the HMM regime-fitting step.

Instead, VIX and CAPE enter only through the optimizer’s reference allocation w_0 . This design separates regime identification from portfolio stabilization. The HMM estimates latent regime probabilities using only the factor-return panel, while the anchor adjusts the optimizer’s reference point under different market-stress or valuation environments. Thus, the anchor is not intended to directly predict factor returns; rather, it provides a sentiment- or valuation-aware stabilization mechanism for the regime-conditional portfolio optimization.

In contrast to treating VIX or CAPE as direct timing predictors, our empirical results suggest that they can be useful when incorporated as optimization anchors. Both variables therefore play an instrumental role in stabilizing portfolio weights and improving out-of-sample allocation performance, even though they are not used as independent forecasting signals.

4.4.1 Three anchor families

We evaluate three anchor families: no anchor, a non-regression anchor, and a regression-based anchor.

No-anchor benchmark. The benchmark specification does not use VIX or CAPE. The regime-conditional optimization is solved without a macro-conditioned warm-start:

$$w_0 = w_{\text{unif}} = \frac{1}{N} \mathbf{1}. \quad (12)$$

Non-regression anchor. The non-regression anchor applies a simple scalar adjustment to the uniform warm-start. Let x_t denote the auxiliary macro variable, either VIX or CAPE. Using only observations available before time t , we compute the expanding-window mean $\bar{x}_t$ and standard deviation $\sigma_{x,t}$. The scalar anchor is

$$s_t = \text{clip} \left(1 - 0.5 \cdot \frac{x_t^{\text{anchor}} - \bar{x}_t}{\sigma_{x,t}}, 0.5, 1.5 \right), \quad w_{0,t} = s_t w_{\text{unif}}. \quad (13)$$

Here x_t^{anchor} is a regime-conditioned macro baseline, defined by the hard or soft mode in Section 4.4.2. The clipping interval $[0.5, 1.5]$ limits the influence of extreme macro adjustments. This family does not estimate factor-specific macro sensitivities; it only scales the uniform warm-start.

Regression-based anchor. The regression-based anchor constructs a factor-specific warm-start vector. In each expanding training window, each factor return is regressed on standardized macro-conditioning variables:

$$r_{i,\tau} = \alpha_i + \beta_i^\top \mathbf{z}_\tau + \varepsilon_{i,\tau}, \quad \tau < t. \quad (14)$$

The current macro state is standardized using the same expanding-window mean and standard deviation, and then clipped at the z-score level:

$$\mathbf{z}_t = \text{clip} \left(\frac{\mathbf{x}_t - \hat{\boldsymbol{\mu}}_{x,t}}{\hat{\boldsymbol{\sigma}}_{x,t}}, -2, 2 \right). \quad (15)$$

The estimated factor sensitivities are converted into a directional warm-start tilt:

$$\tilde{w}_{0,i,t} = \hat{\beta}_{i,t}^\top \mathbf{z}_t, \quad w_{0,i,t} = \frac{\tilde{w}_{0,i,t}}{\sum_{j=1}^N |\tilde{w}_{0,j,t}|}. \quad (16)$$

Unlike the non-regression anchor, which clips the final scalar multiplier to $[0.5, 1.5]$, the regression-based anchor clips the standardized macro input to $[-2, 2]$ before translating macro sensitivity into factor-level tilts. This difference reflects the different role of clipping in the two specifications. In the non-regression anchor, the macro signal only scales a uniform warm-start, so clipping the final scalar directly limits how much the overall starting allocation can expand or contract. In the regression-based anchor, however, the warm-start is constructed from factor-specific sensitivity estimates and then normalized by its ℓ_1 norm. Therefore, clipping the final vector would be less informative; instead, we clip the macro z -score before it interacts with estimated factor betas, preventing extreme VIX or CAPE observations from dominating the tilt direction.

Note that we use standardized macro variables rather than raw levels for three reasons. First, VIX and CAPE are measured on different scales, so using raw values would make regression coefficients difficult to compare across anchors. Second, standardization makes the regression-based anchor directly comparable to the non-regression anchor, which also conditions on deviations from an expanding-window macro baseline. Third, using z -scores improves numerical stability and makes the estimated sensitivities more informative: β_i measures how factor i responds to a one-standard-deviation change in the macro-conditioning variable rather than to an arbitrary one-unit change in its raw level.

4.4.2 Two anchor modes

The hard and soft modes apply to both the non-regression and regression-based anchors. The global mode applies only to the regression-based anchor, because it is defined by a pooled regression over all available observations.

Hard anchor. The hard anchor uses regime-specific information from the currently optimized regime. For the non-regression anchor, the macro baseline is the historical mean of the auxiliary variable within regime r :

$$x_t^{\text{anchor}} = \bar{x}_{r,t}. \quad (17)$$

For the regression-based anchor, factor sensitivities are estimated using only observations assigned to regime r :

$$\hat{\beta}_{i,t}^{\text{hard}} = \hat{\beta}_{i,r,t}. \quad (18)$$

If the regime-specific sample is too small, the method falls back to the global regression.

Soft anchor. The soft anchor averages across regimes using the current HMM posterior probabilities. For the non-regression anchor, the macro baseline is

$$x_t^{\text{anchor}} = \sum_{r=1}^K \gamma_t(r) \bar{x}_{r,t}. \quad (19)$$

For the regression-based anchor, regime-specific regression coefficients are estimated separately and then blended as

$$\hat{\beta}_{i,t}^{\text{soft}} = \sum_{r=1}^K \gamma_t(r) \hat{\beta}_{i,r,t}. \quad (20)$$

This mode smooths the anchor across regime transitions and reduces sensitivity to small changes in the dominant regime label.

Overall, the non-regression anchor provides a simple regime-level macro adjustment, while the regression-based anchor provides a factor-specific sensitivity adjustment. Hard and soft modes determine how regime information enters the anchor, differing in flexibility. All anchor variants are designed to stabilize the optimizer without allowing VIX or CAPE to enter the HMM likelihood or act as standalone return-forecasting signals.

4.5 Evaluation Protocol

We evaluate the framework using a burn-in period followed by a walk-forward deployment period to avoid look-ahead bias. The burn-in period spans 1990–2012 (which is further divided into 1990–1999 anchor regression and 2000–2012 configurations grid-search window), corresponding to 23 years or 276 monthly observations, reflecting the limited historical availability of VIX data. During this period, the HMM and the regime-conditional weight matrix $\{w_r\}_{r=0}^3$ are calibrated as described in Sections 4.1–4.2. The burn-in period is also used to compare different anchor specifications and select the final anchor schemes used in the deployment evaluation.

From January 2013 onward, the pipeline is evaluated under a strict expanding-window protocol. In each month, the HMM is refit using the full history available through month $t - 1$. The regime-conditional weights $\{w_r\}_{r=0}^3$ are reoptimized every two months, while the filtered regime probabilities γ_t and the one-step-ahead predictive regime distribution

$$\hat{\pi}_{t+1|t} = \gamma_t^\top \mathbf{T}$$

are updated monthly. The deployed portfolio weight w_{t+1} is then formed by mixing the regime-conditional solutions according to this predictive distribution, as in Eq. (8). This walk-forward protocol is used for all headline out-of-sample results.

4.6 Final strategies configurations

We report two final configurations representing distinct investor mandates: a long-only *Protective* variant and a long-short *Aggressive* variant. Both share the same regime model but varied in the optimization framework. Hyperparameters are calibrated on the 23-year burn-in period (1990–2012) and held fixed throughout the walk-forward evaluation; sensitivity to alternative specifications across configurations is reported in Section 6 Robustness Checks. Table 3 summarizes the choices.

Table 3: Hyperparameter specification for the main configurations.

Component	Parameter	Long-only	long-short
HMM	Number of regimes	4	4
	Covariance type	Diagonal	Diagonal
	Convergence tolerance	10^{-3}	10^{-3}
	Diagonal strength	0.23	0.23
Anchor	Anchor variable	CAPE	VIX
	Regression anchor	True	True
	Conditioning mode	Hard	Soft
	Anchor strength α	0.01	0.01
Portfolio optimization	Solver	SLSQP	SLSQP
	Covariance estimator	Ledoit–Wolf shrinkage	Ledoit–Wolf shrinkage
	Risk aversion λ	0.4	0.4
	Dominance constraint	Disabled	Disabled
Weight bounds	Minimum weight $w_{\min}$	0	−0.2
	Maximum weight $w_{\max}$	0.5	0.5

5 Results

5.1 Regimes identification

The fitted HMM partitions the 1984–2023² monthly factor panel into four regimes whose mean factor exposures, measured in standardized units, are reported in Table 4. The regime labels follow Botte and Bao (2021), and the four identified states are broadly consistent with their findings.

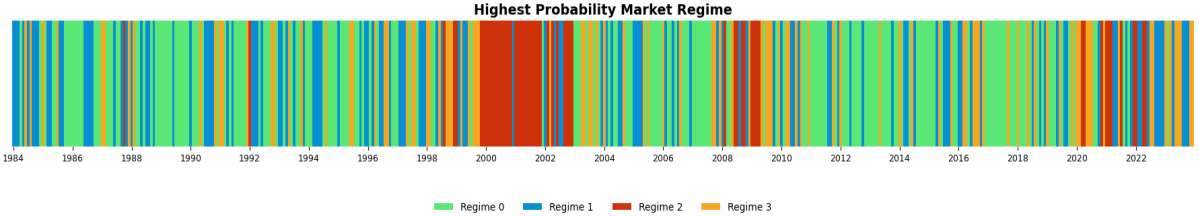


Figure 1: A general filtered regime sequence, 1984–2023. Colored bands indicate the dominant regime each month under the HMM’s filtered posterior. The figure shows that: Regime 0 (Steady State) concentrate during periods of steady growth; Regime 2 (Crisis) concentrates in 2008–09 and 2020 Q1; Regime 3 (Walking on Ice) appears around Regime 2 Crisis.

- **Regime 0 — Steady State.** Modestly positive market (+0.14 SD), with no factor exceeding ± 0.21 SD in magnitude. Profit growth and quality tilt slightly positive; value and investment slightly negative. This is the low-amplitude modal state covering periods of unremarkable equity appreciation without strong factor rotation.
- **Regime 1 — Defensive Drawdown.** Sharply negative market (−0.59 SD) coupled with a clean value/defensive factor pattern: value (+0.47), investment (+0.49), accruals (+0.38),

²The HMM regime-clustering analysis uses the full 1984–2023 factor sample, including 1984–1990. The portfolio strategy formation starts from 1990 because the VIX-based anchor is only available from 1990 onward.

and low-risk (+0.58) all strongly positive, while size (−0.49) and quality (−0.35) lag. The signature is a growth-led equity drawdown in which large-cap value outperforms.

- **Regime 2 — Crisis.** Equally negative market (−0.58 SD) but accompanied by a broad-based factor rally rather than a single-style rotation: seasonality (+0.82), debt issuance (+0.71), quality (+0.45), size (+0.45), investment (+0.46), profitability (+0.30), and value (+0.29) all positive, with momentum still working (+0.15). This is the classical recession environment in which academic factor premia broadly pay off, and we expect substantial overlap with the 2008–09 and 2020 Q1 episodes documented in Botte and Bao (2021).
- **Regime 3 — Walking on Ice.** Strongly positive market (+0.80 SD) with the hallmarks of a post-bear snapback: small caps leading (+0.53), low-risk inverted (−0.73, high-beta names winning), value lagging growth (−0.47), and momentum crashing (−0.24). Typical of high volatility, well-behaved equity markets that tend to occur around Crisis (and Steady State) periods.

Table 4: Per-regime mean factor exposures, in standardized units (full-sample z -scores). The strongest factor per regime is highlighted in bold.

Factor	Regime 0	Regime 1	Regime 2	Regime 3
accruals	−0.14	+0.38	−0.20	−0.08
debt issuance	−0.12	−0.22	+0.71	+0.29
investment	−0.20	+0.49	+0.46	−0.46
low leverage	+0.09	−0.51	−0.08	+0.56
low risk	−0.07	+0.58	+0.26	−0.73
mkt	+0.14	−0.59	−0.58	+0.80
momentum	+0.08	−0.00	+0.15	−0.24
profit growth	+0.15	−0.28	−0.13	+0.19
profitability	−0.01	+0.11	+0.30	−0.28
quality	+0.10	−0.35	+0.45	+0.07
seasonality	−0.04	+0.04	+0.82	−0.28
short-term reversal	−0.10	+0.06	+0.03	+0.14
size	−0.02	−0.49	+0.45	+0.53
value	−0.15	+0.47	+0.29	−0.47

Two structural features stand out from table 4: the market factor cleanly separates the regimes into one strongly bullish state (Regime 3, +0.80 SD) and two bearish states (Regimes 1 and 2, both near −0.6 SD), with Regime 0 occupying a low-amplitude middle ground; and the two bearish states are not redundant — they differ sharply in which factor premia are paying, which is precisely the heterogeneity that motivates the regime-aware allocation.

5.2 Strategy Performance (2013–2023 Walk-Forward Period)

We evaluate the framework over the 2013–2023 out-of-sample window under the walk-forward protocol described in Section 4.5. All model configurations, including the anchor variable, conditioning mode, weight bounds, and optimization hyperparameters, are selected during the burn-in period only as shown in table 3. The walk-forward period is therefore used purely for out-of-sample performance evaluation. Across the configuration grid, we report two main specifications that represent distinct investor objectives: a Sharpe-focused long-only protective strategy and a return-seeking long-short aggressive strategy. Full configuration comparisons are reported in the robustness analysis.

- **Long-only strategy.** This specification uses the CAPE hard-regression anchor with weight bounds $w_{\min} = 0$ and $w_{\max} = 0.5$. No shorting is permitted, so the strategy is designed as the more conservative and implementable allocation.
- **Long-short aggressive strategy.** This specification uses the VIX soft-regression anchor with weight bounds $w_{\min} = -0.2$ and $w_{\max} = 0.5$. Short positions are allowed up to 20% of capital per factor, making this specification a more return-seeking alternative with higher active exposure.

Table 5: Strategies performance 2013–2023.

Metric	Long-only(protective)	Long-short(aggressive)
Sharpe ratio	1.33	1.24
Annualized excess return	5.39%	6.86%
Annualized total return	6.56%	8.03%
Annualized vol.	4.05%	5.54%
Maximum drawdown	-3.64%	-5.78%
Max drawdown length	7.00	12.00
Calmar ratio	1.48	1.19
Win rate	66.41%	66.41%
Turnover(annual) ³	127.74%	308.52%

The long-only specification produces the strongest risk-adjusted performance, with a Sharpe ratio of 1.33, annualized volatility of 4.05%, and maximum drawdown of -3.64% . The aggressive specification accepts higher volatility and turnover in exchange for higher annualized return, delivering 6.86% annualized excess return (annualized total return 8.03%) with a Sharpe ratio of 1.24.

5.3 Benchmark comparison⁴

We compare the aggressive long–short and protective long-only strategies against three passive benchmarks over the 2013–2023 out-of-sample window: SPY (S&P 500), QQQ (Nasdaq-100), and TLT (iShares 20+ Year Treasury Bond ETF).

Figure 2 shows that QQQ and SPY achieve higher cumulative growth during the long post-2013 equity bull market, especially in 2019, 2021, and 2023. However, the selected strategies compound much more smoothly. The long-short aggressive strategy reaches a cumulative growth of 2.34x, while the long-only protective strategy grows to 2.01x with visibly lower volatility. In contrast, TLT performs poorly over the full period, especially during the 2022 bond drawdown.

Figure 3 provides the clearest evidence of the framework’s defensive value. Each series shows the rolling drawdown from its running peak; the strategy’s drawdowns are visibly shallower and shorter-lived than those of the equity and Treasury benchmarks. The underwater plot shows that both model portfolios experience shallower and shorter-lived drawdowns than the equity and Treasury benchmarks. This is especially important during stress episodes such as the 2020 COVID shock and the 2022 joint stock–bond drawdown. While the passive equity benchmarks dominate in strong bull markets, the regime-aware strategies deliver a more stable return path

³Annualized one-way turnover, $TO = 3 \cdot \mathbb{E}[\sum_i |w_{i,t} - w_{i,t-1}|]$, averaged across the 6 bi-monthly rebalance dates per year. The constant 3 = 6/2 combines the bi-monthly frequency and the two-way-to-one-way conversion.

⁴Throughout this section, return-related statistics are reported on an annualized basis. Because the benchmark series are total-return indices, their returns include the risk-free-rate component embedded in index performance.

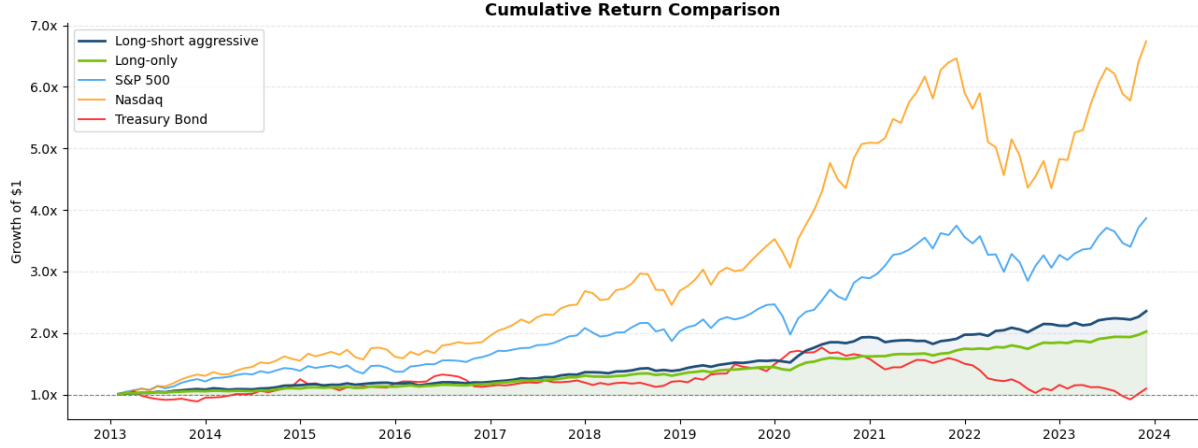


Figure 2: Cumulative returns of strategies versus SPY, QQQ, and TLT, 2013–2023.

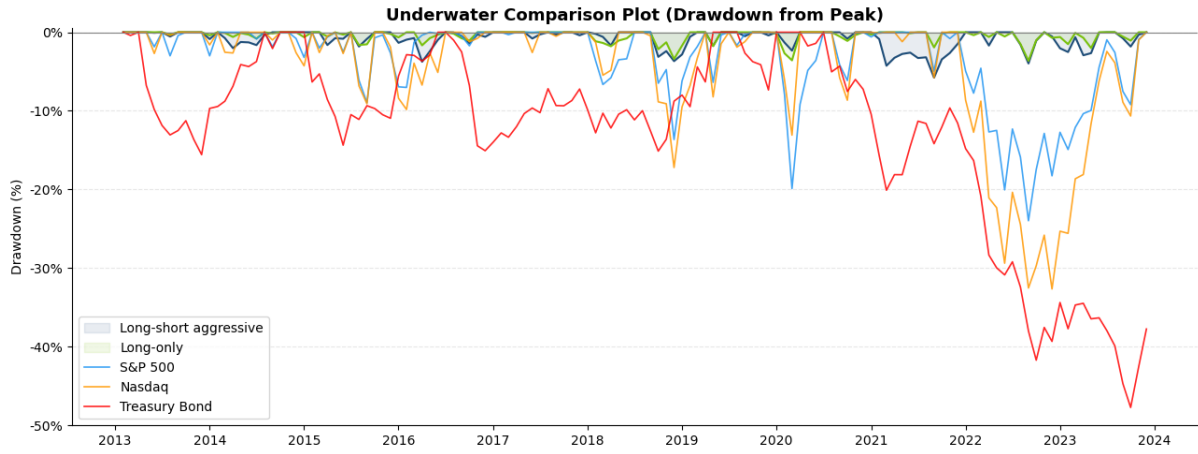


Figure 3: Underwater drawdown comparison, 2013–2023.

with materially better drawdown control. Overall, the benchmark comparison suggests that the main advantage of the framework lies in risk-adjusted performance rather than raw return maximization.

Table 6 reports total-return statistics (excess return plus the contemporaneous risk-free rate) to match the basis of the benchmark series, which is different from the table 5’s statistics (based on excess return). Thus, drawdown, return, and Sharpe figures therefore differ slightly between the two tables by construction.

Table 6: Aggregate performance over the walk-forward period, 2013–2023.

	Ann. Return	Ann. Vol.	Sharpe	Max DD	Calmar	Win Rate	Max DD Length
Long-short aggressive	8.03%	5.54%	1.24	−5.76%	1.19	66.41%	11
Long-only protective	6.56%	4.06%	1.33	−3.35%	1.48	67.94% [†]	5
S&P 500	13.58%	15.06%	0.90	−23.97%	0.57	67.94%	23
Nasdaq	19.21%	18.01%	1.07	−32.65%	0.59	64.12%	23
Treasury Bond	1.79%	13.75%	0.13	−47.72%	0.04	48.85%	41

[†] Adding the contemporaneous risk-free rate changes statistics. Although the long-only strategy and S&P 500 share an identical 67.94% win rate over the 2013–2023 sample, the strategies do not agree month-by-month: 12 months show strategy gains paired with S&P 500 losses, and 12 months show the reverse, with the remaining months agreeing in direction. The match in aggregate win rate is therefore coincidental rather than mechanical.

The aggregate performance statistics strengthen the risk-adjusted case. The long-only protective strategy achieves the highest Sharpe ratio, 1.33, compared with 1.24 for the aggressive strategy, 1.07 for Nasdaq, 0.90 for the S&P 500, and 0.13 for Treasury Bonds. It also delivers the strongest Calmar ratio, 1.48, with a maximum drawdown of only -3.35% . The aggressive strategy produces the higher return among the two model specifications, at 8.03% annualized total return, while maintaining a maximum drawdown of -5.76% . Both model variants therefore substantially reduce downside exposure relative to the benchmark assets, whose maximum drawdowns range from -23.97% for the S&P 500 to -47.72% for Treasury Bonds.

Table 7: Annual returns by calendar year, 2013–2023.

Year	Long–short aggressive	Long–only protective	S&P 500	Nasdaq	Treasury Bond
2013	9.13%	5.90%	25.19%	32.66%	-11.09%
2014	5.38%	4.44%	13.45%	19.12%	27.54%
2015	3.90%	2.45%	1.19%	9.51%	-1.77%
2016	0.11%	3.16%	12.00%	7.11%	1.13%
2017	10.69%	9.81%	21.80%	32.78%	9.24%
2018	4.60%	1.87%	-4.64%	-0.20%	-1.66%
2019	11.77%	10.66%	31.34%	39.06%	14.27%
2020	24.68%	12.26%	18.41%	48.46%	18.30%
2021	-1.20%	5.81%	28.82%	27.49%	-4.63%
2022	12.47%	6.98%	-18.26%	-32.65%	-31.40%
2023	9.90%	10.14%	26.24%	54.84%	2.61%

Table 7 reports annual returns by calendar year. The bolded years highlight the two clearest stress-period validations of the regime-aware thesis. In 2020, the long-short aggressive strategy gained 24.68%, outperforming SPY despite maintaining a more defensive and regime-conditioned allocation. In 2022, during the joint stock–bond drawdown, both model variants performed strongly (12.47% and 6.98%), while SPY, Nasdaq, and TLT declined sharply. These results suggest that the framework’s main value is not simply bull-market participation, but improved resilience during regime transitions and market stress.

5.4 Literature benchmark

As an approximate point of reference, Ilmanen et al. (2021) report the performance of a US-stock multifactor strategy across pre- and post-publication periods. This comparison is relevant because our 2013–2023 evaluation window lies entirely within the modern post-publication factor environment. In their Table II, the US-stock multifactor Sharpe ratio declines from 1.83 in the original-sample period to 0.78 post-publication, highlighting the deterioration of unconditional factor premia after discovery. Against this benchmark, our long-only protective strategy (CAPE hard-regression anchor) achieves a Sharpe ratio of 1.33 and the long-short aggressive strategy (VIX soft-regression anchor) achieves 1.24 over 2013–2023. We emphasize that this is not a controlled comparison: the factor universe, construction methodology, and evaluation period differ between the two studies, and the Ilmanen et al. multifactor post-period is constructed from a subperiod split rather than a single clean post-publication date. We therefore treat it only as a coarse indication that the regime-aware framework remains economically meaningful in an environment where static factor premia have become substantially harder to exploit, rather than as a strict replication.

5.5 Drawdown Analysis

Tables 8 and 9 summarize the largest drawdown episodes for the two selected leakage-free specifications. The long-short aggressive strategy uses the VIX soft-regression anchor, while the

long-only strategy uses the CAPE hard-regression anchor. The aggressive strategy experiences a maximum drawdown of -5.78% , beginning in February 2021, reaching its trough in September 2021, and recovering by January 2022. This episode lasts 12 months with a five-month recovery period, reflecting the higher risk budget, shorting flexibility, and greater turnover allowed by the aggressive specification. Its next largest drawdowns occur in August–November 2022 and January–July 2016, with depths of -3.97% and -3.75% , respectively.

Table 8: Top 10 drawdown episodes: long-short aggressive strategy

Start	Trough	Recovery	Depth	Duration (Months)	Recovery (Months)
2021-02	2021-09	2022-01	-5.78%	12	5
2022-08	2022-09	2022-11	-3.97%	4	3
2016-01	2016-04	2016-07	-3.75%	7	4
2018-10	2018-12	2019-03	-3.68%	6	4
2022-12	2023-04	2023-07	-2.93%	8	4
2020-02	2020-03	2020-04	-2.34%	3	2
2014-03	2014-04	2014-09	-2.02%	7	6
2023-09	2023-10	2023-12	-1.82%	4	3
2015-08	2015-08	2015-10	-1.82%	3	3
2019-05	2019-05	2019-06	-1.74%	2	2

The long-only strategy displays materially stronger drawdown control. Its worst drawdown is -3.64% , occurring during the 2022 stock–bond drawdown from August 2022 to September 2022, with recovery by November 2022. The second largest drawdown is the COVID episode in early 2020, with a depth of -3.58% and recovery by April 2020. Although the long-only strategy still experiences losses during major market stress periods, the losses are shallower and generally recover faster than those of the aggressive strategy.

Table 9: Top 10 drawdown episodes: long-only strategy

Start	Trough	Recovery	Depth	Duration (Months)	Recovery (Months)
2022-08	2022-09	2022-11	-3.64%	4	3
2020-01	2020-03	2020-04	-3.58%	4	2
2018-09	2018-12	2019-02	-3.40%	6	3
2022-12	2023-05	2023-06	-2.01%	7	2
2021-09	2021-09	2021-11	-1.93%	3	3
2018-02	2018-04	2018-07	-1.78%	6	4
2019-05	2019-05	2019-06	-1.76%	2	2
2016-04	2016-04	2016-07	-1.67%	4	4
2015-08	2015-08	2015-10	-1.62%	3	3
2020-09	2020-10	2020-12	-1.10%	4	3

Overall, the drawdown evidence confirms the trade-off between the two specifications. The aggressive strategy delivers higher annualized return but accepts deeper and more persistent interim losses, with a worst drawdown of -5.78% and a longest drawdown episode lasting 12 months. The long-only strategy sacrifices some return but provides superior downside control, with a maximum drawdown of only -3.64% over the full 2013–2023 out-of-sample window. Compared with passive benchmarks such as the S&P 500, Nasdaq, and Treasury Bonds, both regime-aware strategies experience substantially smaller drawdowns and faster recoveries. This suggests that the main advantage of the framework lies in stabilizing performance during stressed or transitional regimes rather than simply maximizing raw return.

5.6 Factor exposure analysis

We next examine the realized factor exposures of the two strategies. The exposure analysis helps clarify where the performance comes from and whether the strategies behave like diversified factor allocations or concentrated bets on a small number of premia.

Long-short aggressive strategy. The long-short aggressive specification is more active and uses both positive and negative factor tilts. Its largest average long exposures are to quality, market, low leverage, momentum, and profitability, with average weights of 28.45%, 26.74%, 25.58%, 16.59%, and 12.66%, respectively. At the same time, the strategy maintains persistent short or underweight positions in accruals, short-term reversal, size, seasonality, and profit growth, with average weights of -7.25% , -4.72% , -2.30% , -3.88% , and -2.35% , respectively. This suggests that the aggressive portfolio is not simply a levered market allocation; it combines a market component with quality, low-leverage, and momentum tilts while using short positions to reduce exposure to weaker or unstable factors. Figure 4 shows that the aggressive strategy is highly dynamic. Most factors are active in nearly every month, and the heatmap displays frequent reallocations across the sample. This flexibility contributes to higher return generation but also increases turnover. The average bi-monthly turnover is approximately 51.4%, and the average concentration index is 0.33, indicating a more concentrated and actively managed allocation than the long-only variant. This is consistent with the strategy’s higher annualized return but also its larger drawdown and higher implementation burden.

Table 10: Factor exposure summary for the long-short aggressive strategy, 2013–2023.

Factor	Avg Weight	Std Weight	Min Weight	Max Weight	Active (%)
accruals	-7.25%	5.43%	-15.27%	5.97%	98.48%
debt issuance	0.82%	3.92%	-10.10%	7.79%	98.48%
investment	3.31%	5.39%	-7.67%	21.82%	98.48%
low leverage	25.58%	6.32%	7.68%	37.67%	100.00%
low risk	3.60%	5.69%	-7.67%	21.82%	99.24%
mkt	26.74%	5.42%	8.18%	37.67%	100.00%
momentum	16.59%	7.03%	-0.64%	34.43%	100.00%
profit growth	-2.35%	5.52%	-12.05%	11.72%	96.97%
profitability	12.66%	7.85%	-0.02%	32.68%	98.48%
quality	28.45%	4.96%	19.75%	41.29%	100.00%
seasonality	-3.88%	4.25%	-15.39%	3.22%	97.73%
ST reversal	-4.72%	4.30%	-12.44%	5.16%	97.73%
size	-2.30%	11.00%	-18.29%	26.37%	97.73%
value	2.76%	5.62%	-7.67%	21.79%	97.73%

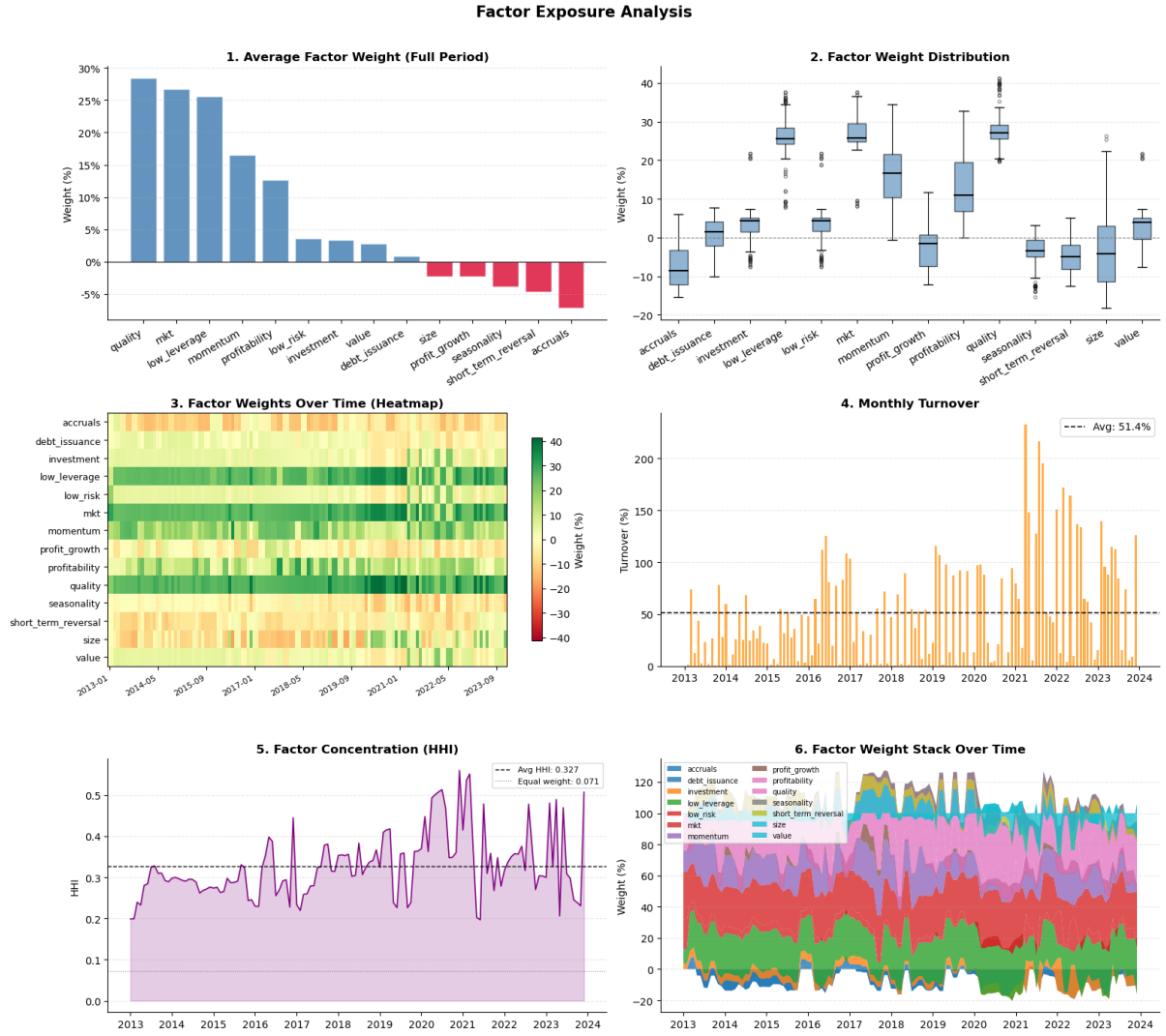


Figure 4: Long-short strategy factor exposure.

Long-only protective strategy. The long-only protective specification is more stable and concentrated in a smaller set of persistent positive exposures. Its largest average allocations are to market, quality, low risk, low leverage, value, and investment, with average weights of 28.62%, 25.68%, 16.68%, 12.27%, 6.88%, and 4.83%, respectively. Unlike the aggressive version, the long-only strategy assigns almost no average exposure to accruals, profit growth, short-term reversal, and size, and only marginal exposure to debt issuance, seasonality, momentum, and profitability. This reflects the effect of the no-shorting constraint: instead of expressing negative views through short positions, the optimizer simply avoids unattractive factors.

Table 11: Factor exposure summary for the long-only strategy, 2013–2023.

Factor	Avg Weight	Std Weight	Min Weight	Max Weight	Active (%)
accruals	0.00%	0.00%	0.00%	0.00%	0.00%
debt issuance	0.49%	0.57%	0.00%	1.78%	59.85%
investment	4.83%	1.96%	0.91%	8.97%	100.00%
low leverage	12.27%	9.21%	0.61%	30.84%	100.00%
low risk	16.68%	3.86%	8.81%	29.87%	100.00%
mkt	28.62%	6.98%	11.48%	38.04%	100.00%
momentum	3.02%	1.99%	0.55%	13.77%	100.00%
profit growth	0.00%	0.00%	0.00%	0.01%	0.00%
profitability	1.52%	0.92%	0.00%	4.01%	84.85%
quality	25.68%	8.07%	2.47%	34.26%	100.00%
seasonality	0.01%	0.02%	0.00%	0.12%	3.03%
ST reversal	0.00%	0.00%	0.00%	0.00%	0.00%
size	0.00%	0.00%	0.00%	0.00%	0.00%
value	6.88%	3.76%	1.53%	22.56%	100.00%



Figure 5: Long-only strategy factor exposure. Same graph format as Figure 4.

Figure 5 confirms that the long-only strategy is much smoother. The heatmap and stacked-weight plot show persistent allocations to market, quality, low risk, low leverage, and investment, with only moderate time variation. Average monthly turnover falls to 21.3%, more than half that of the aggressive strategy, and the average concentration index decreases to 0.22. This lower turnover and lower concentration help explain the strategy’s stronger drawdown control, lower annualized volatility, and higher Sharpe ratio.

5.7 Transaction Costs, Implementation Frictions, and Net Performance

A practical evaluation of the regime-aware allocation framework must account for implementation costs, which can materially erode the gross alpha documented in factor backtests. Building on the factor exposure analysis in Section 5.6, this section provides a simple weight-based estimate of trading frictions and net-of-cost performance. The 14-factor universe exhibits substantial heterogeneity in implementation difficulty: some factors can be approximated using liquid market or factor ETFs, while others require stock-level replication to obtain cleaner academic factor exposure. In addition, the regime-conditioning mechanism introduces further trading demand because portfolio weights vary over time with the inferred market regime. The factors are divided into two implementation groups:

- **ETF-source factors:** market, low risk, momentum, profitability, quality, size, and value. These factors have relatively direct ETF proxies through broad-market and smart-beta products.
- **Equity-replicated factors:** accruals, debt issuance, investment, low leverage, profit growth, seasonality, and short-term reversal. These factors either lack clean standalone ETFs or require firm-level accounting and return-sorting rules to match the academic factor definition.

For the long-only protective strategy, the average allocation is concentrated in ETF-source factors. Based on Table 11, approximately 82.40% of the average portfolio weight can be implemented using ETF-source factors, while the remaining 17.60% requires equity-level replication. For the long-short aggressive strategy, negative positions make gross exposure more informative than net weight. Using absolute average factor exposures, approximately 66.02% of the gross exposure is ETF-source exposure, while 33.98% is equity-replicated exposure. Therefore, the long-short strategy is expected to face larger implementation frictions because it combines higher turnover, larger exposure to harder-to-replicate factors, and short-position costs.

For simplicity, the following transaction-cost estimate uses a first-order turnover decomposition. Specifically, ETF-source and equity-replicated turnover are approximated by multiplying total annual turnover by the average ETF-source and equity-replicated portfolio shares. A more precise real-world implementation would compute turnover directly from the realized monthly rebalancing path and deduct costs from monthly returns before recomputing net performance metrics.

Long-only protective strategy transaction-cost estimation. For the long-only protective strategy, annual turnover is 127.74%. Since approximately 82.40% of the average allocation is ETF-source exposure and 17.60% is equity-replicated exposure, the implied annual ETF-source turnover is approximately

$$127.74\% \times 82.40\% = 105.26\%,$$

while the implied annual equity-replication turnover is approximately

$$127.74\% \times 17.60\% = 22.48\%.$$

Assuming a base-case trading cost of 5 bps for ETF trades and 30 bps for equity-replicated trades, the annual cost drag is

$$1.0526 \times 0.05\% + 0.2248 \times 0.30\% = 0.12\%.$$

Therefore, the long-only strategy's annualized excess return decreases from 5.39% to approximately 5.27%. Using the annualized volatility of 4.05%, the approximate net Sharpe ratio becomes

$$\frac{5.39\% - 0.12\%}{4.05\%} = 1.30.$$

Thus, estimated implementation costs reduce the long-only strategy's performance only modestly. The main reason is that most of the strategy is implementable through liquid ETF-source sleeves, while the higher-cost equity-replicated component represents a relatively small share of the average allocation. Even after transaction-cost adjustment, the long-only protective strategy remains economically meaningful and retains most of its gross Sharpe performance.

Long-short aggressive strategy transaction-cost estimation. The long-short aggressive strategy has substantially higher annual turnover of 308.52%. Because the strategy includes negative factor positions, implementation shares are computed using absolute average exposures rather than net average weights. Based on Table 10, ETF-source factors account for approximately 66.02% of gross exposure, while equity-replicated factors account for approximately 33.98%. This implies annual ETF-source turnover of approximately

$$308.52\% \times 66.02\% = 203.70\%,$$

and annual equity-replication turnover of approximately

$$308.52\% \times 33.98\% = 104.82\%.$$

Using the same base-case assumptions of 5 bps for ETF trades and 30 bps for equity-replicated trades, the annual transaction-cost drag is

$$2.0370 \times 0.05\% + 1.0482 \times 0.30\% = 0.42\%.$$

Therefore, the long-short strategy's annualized excess return decreases from 6.86% to approximately 6.44%. Using annualized volatility of 5.54%, the approximate net Sharpe ratio becomes

$$\frac{6.86\% - 0.42\%}{5.54\%} = 1.16.$$

This estimate likely remains optimistic because it does not fully incorporate short-sale borrow fees, short-rebate frictions, financing costs, or the additional operational complexity of maintaining short factor exposures. The long-short strategy has approximately 20.50% average short exposure based on negative average weights, so even modest annual shorting frictions would further reduce net performance.

Table 12: Estimated net-of-cost performance under base-case implementation costs

Metric	Long-only protective	Long-short aggressive
Gross annualized excess return	5.39%	6.86%
Annual turnover	127.74%	308.52%
ETF-source turnover	105.26%	203.70%
Equity-replicated turnover	22.48%	104.82%
Estimated annual cost drag	0.12%	0.42%
Net annualized excess return	5.27%	6.44%
Approx. net Sharpe ratio	1.30	1.16

Overall, the transaction-cost analysis strengthens the distinction between the two strategies. The long-short aggressive strategy generates higher gross annualized excess return, but it is more vulnerable to implementation frictions because of its higher turnover, larger equity-replication share, and shorting costs. In contrast, the long-only protective strategy is more implementation-friendly: it has lower turnover, a larger ETF-source allocation, and no short-sale friction. Therefore, after estimated transaction costs, the long-only strategy remains the more practical and robust implementation pathway.

6 Robustness Checks

6.1 Regime identification

Following Botte and Bao (2021), we treat latent market regimes as mixture components estimated from the factor-return panel. Since both the GMM and HMM specifications are likelihood-based Gaussian models, the number of regimes is selected by comparing out-of-sample likelihood performance and information criteria across candidate values of K (number of regimes).

To validate the regime choice, we implement an expanding-window train–test procedure over the full sample from January 1984 to December 2023. The initial training window contains 12 years of monthly observations, and each test window contains four years. For each candidate number of regimes, the model is re-estimated across the expanding windows and evaluated using average test log-likelihood, AIC, and BIC. Higher test log-likelihood indicates better out-of-sample fit, while lower AIC and BIC indicate better penalized in-sample fit.

Table 13 reports the model-selection results. In both the GMM and HMM checks, the four-regime specification achieves the highest average out-of-sample test log-likelihood. Although AIC and BIC sometimes continue to favor larger values of K , these criteria mechanically reward additional flexibility in high-dimensional likelihood models and may select regimes that are less stable or less economically interpretable. We therefore choose $K = 4$ as the final specification because it maximizes out-of-sample likelihood in both validation exercises, aligns with the four-regime structure in Botte and Bao (2021), and provides interpretable market-state classifications without excessive fragmentation.

Table 13: Model selection statistics by number of clusters under two comparison methods.

Panel	Model	Avg. Test LL	Avg. AIC	Avg. BIC
Panel A: GMM				
	cluster2	-1326.91	8016.27	9017.77
	cluster3	-1351.58	8398.09	8702.12
	cluster4	-1285.37	8107.29	8513.83
	cluster5	-1335.67	7898.53	8407.60
	cluster6	-1312.75	7738.28	8349.87
	cluster7	-1401.08	7639.64	8353.75
	cluster8	-1438.47	7536.38	8353.01
Panel B: HMM				
	cluster2	-1327.34	8765.20	8973.78
	cluster3	-1355.31	8326.78	8652.02
	cluster4	-1287.72	8038.18	8487.14
	cluster5	-1422.52	7838.97	8418.74
	cluster6	-1366.89	7690.58	8408.22

Although the plain model-selection results suggest that the GMM achieves slightly stronger clustering fit, we ultimately select the HMM as the final regime-identification model. For the same four-regime specification, the GMM attains a marginally higher average test log-likelihood

than the HMM (-1285.37 versus -1287.72), corresponding to only a small relative reduction of approximately 1.83×10^{-3} . This difference is economically minor relative to the practical advantage of the HMM.

The key limitation of the GMM is that it treats each observation as independent and provides no transition mechanism. In a real-time walk-forward allocation setting, a GMM-based strategy can only approximate the next-period regime by assuming that tomorrow’s regime remains the same as today’s dominant cluster. By contrast, the HMM explicitly estimates regime persistence through the transition matrix and allows posterior probabilities to be projected one step ahead. This Markov structure makes the HMM more suitable for forward-looking portfolio construction, even though the four-cluster GMM achieves a slightly higher test likelihood. We therefore choose the $K = 4$ HMM as the final regime model.

6.2 Comprehensive Burn-In Configuration Selection

The burn-in period serves as the validation stage for both portfolio hyperparameters and anchor-related design choices. Specifically, we search over covariance estimators, risk-aversion parameters, weight bounds, and anchor specifications, including VIX-based, CAPE-based, and unanchored alternatives. All selection is performed using only burn-in data, so no information from the 2013–2023 walk-forward evaluation period enters the configuration choice.

The unanchored specification is included in the grid as an important benchmark. However, it is not treated as the final anchored model because the methodological contribution of this paper is to evaluate whether macro-conditioned anchors can stabilize regime-aware allocation. Therefore, the no-anchor case is retained as a diagnostic benchmark and is reported later in the robustness analysis, while the main specifications are selected from the anchored candidates.

Long-short aggressive specification. For the long-short grid, the highest raw burn-in score is achieved by the unanchored specification with Ledoit–Wolf covariance, $\lambda = 0.3$, $w_{\min} = -0.2$, and $w_{\max} = 0.5$. This confirms that the no-anchor benchmark is competitive in the burn-in period. Nevertheless, since the unanchored model is reserved as the benchmark case, the selected anchored long-short specification is the best-performing anchored alternative. Among the anchored candidates, the VIX soft-regression anchor delivers the strongest burn-in performance, with annualized return of 10.43%, annualized volatility of 6.46%, maximum drawdown of -5.10% , and Calmar ratio of 2.04. This supports the use of VIX as the anchor variable for the aggressive long-short strategy.

Table 14: Top burn-in configurations for the long-short specification. The no-anchor case is retained as a benchmark; the selected anchored specification is VIX soft-regression.

Cov.	λ	$w_{\min}$	$w_{\max}$	Anchor	Sharpe	Ann. Ret.	Ann. Vol.	Max DD	Calmar
L–W	0.3	−0.2	0.5	None	1.798	10.09%	5.61%	−3.70%	2.73
L–W	0.4	−0.2	0.5	VIX soft	1.613	10.43%	6.46%	−5.10%	2.04
Sample	0.3	0.0	0.5	CAPE hard	1.714	7.14%	4.17%	−7.01%	1.02
L–W	0.3	0.0	0.5	None	1.691	6.94%	4.11%	−7.56%	0.92
Exp	0.4	0.0	0.5	VIX	1.690	6.94%	4.11%	−7.56%	0.92
Sample	0.2	0.0	0.5	None	1.708	7.02%	4.11%	−7.20%	0.97
L–W	0.4	0.0	0.5	None	1.707	7.02%	4.11%	−7.20%	0.97
L–W	0.4	0.0	0.5	CAPE hard	1.718	7.10%	4.14%	−7.00%	1.02
Exp	0.4	0.0	0.5	CAPE hard	1.694	7.04%	4.15%	−7.23%	0.97
Sample	0.4	0.0	0.5	None	1.708	7.01%	4.11%	−7.18%	0.98

Long-only specification. For the long-only grid, the strongest anchored specification uses the CAPE hard-regression anchor with Ledoit–Wolf covariance, $\lambda = 0.4$, $w_{\min} = 0$, and $w_{\max} = 0.5$.

This configuration achieves a burn-in Sharpe ratio of 1.72, annualized return of 7.10%, annualized volatility of 4.14%, and turnover of 16.77%. The result supports the use of a valuation-conditioned anchor for the more conservative long-only strategy. Unlike the aggressive specification, which benefits from the more responsive VIX soft anchor, the long-only strategy is better served by the smoother CAPE hard anchor.

Table 15: Top burn-in configurations for the long-only specification.

Cov.	λ	$w_{\min}$	$w_{\max}$	Anchor	Sharpe	Ann. Ret.	Ann. Vol.	Max DD	Turnover
L-W	0.4	0.0	0.5	CAPE hard	1.718	7.10%	4.14%	-7.00%	16.77%
Sample	0.3	0.0	0.5	CAPE hard	1.714	7.14%	4.17%	-7.01%	16.69%
Sample	0.2	0.0	0.5	None	1.708	7.02%	4.11%	-7.20%	16.04%
L-W	0.4	0.0	0.5	None	1.707	7.02%	4.11%	-7.20%	16.04%
Sample	0.2	0.0	0.5	VIX soft	1.707	6.99%	4.09%	-7.17%	16.05%
Sample	0.2	0.0	0.5	CAPE hard	1.713	7.11%	4.15%	-7.00%	17.05%
Exp	0.2	0.0	0.5	CAPE hard	1.713	7.11%	4.15%	-7.00%	17.05%
L-W	0.2	0.0	0.5	CAPE hard	1.714	7.09%	4.14%	-6.99%	17.11%
Exp	0.4	0.0	0.5	CAPE hard	1.714	7.08%	4.13%	-6.98%	17.13%

Overall, the burn-in selection supports different anchor choices for the two investor objectives. The aggressive long-short strategy uses the VIX soft-regression anchor because it responds more flexibly to market-stress changes, while the long-only strategy uses the CAPE hard-regression anchor because it provides a more stable valuation-conditioned reference point. The unanchored specification remains an important benchmark, but it is not selected as the main model because it does not test the paper’s central anchored-allocation mechanism.

6.3 HMM and GMM performance compare

To evaluate the incremental contribution of the HMM, we compare the final anchored specifications against an otherwise identical GMM benchmark.

Table 16: Walk-forward comparison between HMM and GMM specifications

Strategy	Model	Sharpe	Ann. Return	Ann. Vol	Max DD	Calmar	Turnover
Long-short aggressive	HMM	1.24	6.86%	5.54%	-5.78%	1.19	51.42%
Long-short aggressive	GMM	0.30	3.83%	12.95%	-26.67%	0.14	244.76%
Long-only protective	HMM	1.33	5.39%	4.05%	-3.64%	1.48	21.29%
Long-only protective	GMM	0.44	3.05%	6.92%	-12.63%	0.24	106.30%

Table 16 compares the final HMM-based strategies with otherwise comparable GMM-based alternatives. The results show that the HMM specification consistently dominates the static GMM benchmark across both the long-short aggressive and long-only protective strategies. In the long-short case, the HMM improves the Sharpe ratio from 0.30 to 1.24 while reducing annualized volatility from 12.95% to 5.54% and maximum drawdown from -26.67% to -5.78%. Turnover also falls substantially, from 244.76% under the GMM to 51.42% under the HMM.

The same pattern holds for the long-only protective strategy. The HMM increases the Sharpe ratio from 0.44 to 1.33, reduces maximum drawdown from -12.63% to -3.64%, and lowers turnover from 106.30% to 21.29%. These results suggest that the dynamic transition structure of the HMM provides more stable and implementable regime signals than the static GMM. Although the GMM can classify observations into latent regimes, it does not model regime persistence or provide a transition-based forecast of next-period regime probabilities. This limitation appears to translate into less stable portfolio allocations, higher turnover, and weaker out-of-sample risk-adjusted performance.

Table 17: Burn-in performance comparison between GMM and HMM specifications

Model	Selection	Cov.	λ	α	$w_{\min}$	$w_{\max}$	Anchor	Sharpe	Ann. Ret.	Ann. Vol.	Max DD	Calmar	Bi-monthly turnover
GMM	Burn-in best 1	L-W	0.2	0.01	-0.2	0.5	VIX-reg-soft	2.03	39.04%	19.23%	-37.09%	1.05	248.79%
GMM	Burn-in best 2	SampleCov	0.4	0.01	0.0	0.5	VIX-reg-hard	2.86	35.99%	12.58%	-13.11%	2.74	249.90%
HMM	Burn-in best 1	L-W	0.4	0.01	0.0	0.5	CAPE-reg-hard	1.72	7.10%	4.14%	-7.00%	1.02	16.77%
HMM	Burn-in best 2	L-W	0.4	0.01	-0.2	0.5	VIX-reg-soft	1.61	10.43%	6.46%	-5.10%	2.04	31.35%

Table 17 reports the best burn-in configurations selected under the GMM and HMM specifications. The GMM-based strategies achieve substantially higher burn-in Sharpe ratios and annualized returns than the HMM-based strategies. However, as of table 16 indicates, this burn-in advantage does not carry over to the strict walk-forward deployment period. In the 2013–2023 walk-forward evaluation, the GMM Sharpe ratios fall to 0.30 for the long-short aggressive strategy and 0.44 for the long-only protective strategy.

Such asymmetric burn-in/walk-forward gap has a direct mechanical explanation. During burn-in, both models benefit from in-sample regime inference, since the regime model is fit once on the full burn-in window and regime probabilities are computed from that single fit. This gives the GMM an especially favorable setting, because regime labels are inferred using the full burn-in sample. In deployment, however, this in-sample information is unavailable. The HMM can still project regime probabilities forward through its transition matrix, using $\gamma_t \mathbf{T}$, which smooths allocation changes over time. The GMM has no transition mechanism, so deployment relies more directly on the current inferred cluster and is therefore more sensitive to noisy regime flips.

We interpret this result as empirical support for using a Markov-persistent regime model in a deployed allocation framework. Although the GMM can generate stronger in-sample burn-in performance, its advantage appears to rely heavily on in-sample regime inference. The HMM’s transition matrix acts as an implicit regularizer, stabilizing regime probabilities under strict walk-forward evaluation and producing lower turnover, smaller drawdowns, and stronger out-of-sample risk-adjusted performance. This distinction also aligns with real-world implementation. In deployment, the portfolio does not simply assume that next period’s regime weights are equal to today’s regime assignment. Instead, the HMM uses the estimated transition matrix to project current filtered probabilities one step ahead, $\gamma_t \mathbf{T}$, and forms portfolio weights based on this forward-looking regime distribution. This makes the allocation rule more realistic and smoother than a GMM-based rule that effectively relies on the current dominant cluster as a proxy for next period’s regime.

6.4 Static equal-weight benchmark

As a final robustness check, we compare the regime-aware strategies with a static equal-weight benchmark. The benchmark assigns a constant 1/14 weight to each of the fourteen factor return series over the 2013–2023 walk-forward period. We emphasize that this is an equal-weight portfolio over factor *return series*, not an investable equal-dollar stock portfolio. Since most of the factor series are themselves long–short tercile portfolios with relatively low standalone volatility, the resulting benchmark is naturally low-volatility and risk-light.

This benchmark removes all dynamic components of the proposed framework. It contains no HMM regime inference, no covariance estimation, no turnover-aware optimization, and no VIX or CAPE anchor. Because the factor universe is held fixed, the comparison isolates whether the regime-aware allocation procedure adds value beyond naive diversification across the same factors.

Table 18: Static equal-weight ($1/N$) benchmark versus the regime-aware strategies, 2013–2023. Return-related statistics are reported on an excess-return basis to match Table 12 and the strategy performance reported in Section 5.2.

Metric	Static equal-weight	Long-only protective	Long-short aggressive
Sharpe ratio	1.3206	1.33	1.24
Annualized excess return	1.92%	5.39%	6.86%
Annualized volatility	1.45%	4.05%	5.54%
Maximum drawdown	−2.32%	−3.64%	−5.78%
Calmar ratio	0.8246	1.48	1.19
Turnover (annual)	0.00%	127.74%	308.52%

Table 18 shows that the static equal-weight benchmark is highly competitive on raw Sharpe ratio. Its Sharpe ratio of 1.3206 is close to the long-only strategy’s 1.33 and above the long-short strategy’s 1.24. This result is not surprising: diversified combinations of factor premia often perform well because individual factors have imperfect correlations and because the equal-weight benchmark is extremely low-volatility. Therefore, the relevant question is not whether the proposed framework mechanically dominates naive diversification on Sharpe ratio, but whether it delivers a more useful return and drawdown profile after accounting for the scale of risk taken.

The comparison suggests that the regime-aware strategies add value mainly through return scaling and downside control. The equal-weight benchmark earns only 1.92% annualized excess return at 1.45% volatility. In contrast, the long-only and long-short strategies earn 5.39% and 6.86% annualized excess return, respectively, while maintaining controlled maximum drawdowns of −3.64% and −5.78%. The long-only strategy also improves the Calmar ratio from 0.8246 to 1.48, indicating stronger return generation per unit of maximum drawdown.

Table 19: Top drawdown episodes: static equal-weight ($1/N$) benchmark, 2013–2023.

Start	Trough	Recovery	Depth	Duration (Months)	Recovery (Months)
2019-11	2021-01	2021-05	−2.32%	19	5
2023-01	2023-05	2023-11	−1.15%	11	7
2022-08	2022-09	2022-10	−1.05%	3	2
2018-09	2019-01	2019-08	−0.89%	12	8
2016-04	2016-04	2016-06	−0.76%	3	3

Table 19 reports the largest drawdown episodes of the static equal-weight benchmark. In absolute terms, these drawdowns are shallow, with a maximum drawdown of only −2.32%. However, this reflects the benchmark’s very low volatility rather than superior downside management. A portfolio with 1.45% annualized volatility cannot draw down much in absolute terms unless it is levered.

A leverage-equivalent comparison is therefore more informative. Matching the long-only strategy’s 5.39% annualized excess return would require leveraging the equal-weight benchmark by approximately $5.39/1.92 \approx 2.8$ times. Matching the long-short strategy’s 6.86% annualized excess return would require approximately $6.86/1.92 \approx 3.6$ times leverage. Applying these leverage multiples to the equal-weight benchmark’s maximum drawdown using the geometric scaling

$$1 - (1 - D)^L$$

implies leverage-equivalent maximum drawdowns of approximately −6.4% and −8.0%, respectively. By comparison, the long-only and long-short strategies realize maximum drawdowns of only −3.64% and −5.78%. Thus, on a return-matched basis, both regime-aware strategies draw down less than a scaled static benchmark.

This difference reflects the value of state-dependent allocation. Uniform leverage scales gains and losses symmetrically, while the regime-aware strategies adjust exposures as inferred market conditions change. The equal-weight benchmark’s longest underwater episode begins in late 2019 and lasts 19 months, spanning the COVID shock and its aftermath. The regime-aware strategies, by contrast, experience shorter and shallower drawdowns during the main stress episodes documented in Section 5.5.

The comparison also clarifies the role of turnover. The equal-weight benchmark has zero turnover by construction, while the regime-aware strategies trade actively and therefore incur implementation costs. The net-of-cost analysis in Section 5.7 shows that the dynamic strategies remain economically meaningful after estimated transaction costs. We therefore interpret the equal-weight comparison not as evidence of large Sharpe-ratio dominance, but as evidence that the framework’s value lies in higher deployable return, better leverage-equivalent drawdown control, and state-dependent risk reduction.

6.5 Anchor versus no-anchor comparison

To evaluate the incremental contribution of the anchor mechanism, we compare the final anchored specifications against an otherwise identical unanchored benchmark.

Long-short anchor vs non-anchor. Table 20 compares the selected VIX soft-regression anchor against an otherwise comparable no-anchor long-short benchmark. The anchored specification improves the annualized excess return from 6.15% to 6.86% and slightly increases the Sharpe ratio from 1.22 to 1.24. This suggests that the anchor adds value primarily by improving the return-seeking profile of the aggressive strategy.

Table 20: Long-short strategy: anchored specification versus no-anchor benchmark.

Metric	No-anchor benchmark	VIX soft-regression anchor
Sharpe ratio	1.2201	1.2388
Annualized excess return	6.15%	6.86%
Annualized volatility	5.04%	5.54%
Maximum drawdown	−4.50%	−5.78%
Calmar ratio	1.3663	1.1870
Turnover per rebalance	41.51%	51.42%
Annualized turnover [†]	249.06%	308.52%

[†] Annualized turnover is computed by multiplying the bi-monthly turnover by six.

However, the improvement is not costless. The anchored strategy has higher volatility, a deeper maximum drawdown, and higher turnover than the no-anchor benchmark. Therefore, the anchor should not be interpreted as uniformly reducing risk in the aggressive specification. Instead, it shifts the strategy toward a higher-return, more active allocation that is consistent with the aggressive strategy’s goal, while preserving a modestly stronger risk-adjusted return.

Long-only anchor vs non-anchor. Table 21 compares the selected CAPE hard-regression anchor against the no-anchor long-only benchmark. The anchored specification improves the Sharpe ratio from 1.16 to 1.33 while keeping the annualized excess return nearly unchanged. The main benefit comes from risk reduction: annualized volatility declines from 4.61% to 4.05%, maximum drawdown improves from −6.43% to −3.64%, and the Calmar ratio rises from 0.83 to 1.48.

Table 21: Long-only strategy: anchored specification versus no-anchor benchmark.

Metric	No-anchor benchmark	CAPE hard-regression anchor
Sharpe ratio	1.1602	1.3306
Annualized excess return	5.35%	5.39%
Annualized volatility	4.61%	4.05%
Maximum drawdown	−6.43%	−3.64%
Calmar ratio	0.8317	1.4822
Turnover per rebalance	19.61%	21.29%
Annualized turnover [†]	117.66%	127.74%

[†] Annualized turnover is computed by multiplying the bi-monthly turnover by six.

This suggests that, for the long-only strategy, the anchor primarily improves downside control and risk-adjusted performance rather than increasing raw return. The improvement comes with only a modest increase in turnover, from 19.61% to 21.29% per rebalance, demonstrating the overall effectiveness of anchoring.

7 Discussion

The results should be interpreted through the paper’s four main contributions. First, the framework extends the static GMM-style regime characterization of Botte and Bao (2021) from Two Sigma into a Markov-persistent HMM, so that regimes are not treated as independent monthly clusters but as latent states with estimated transition dynamics. Second, the framework moves from regime description to allocation: the estimated regime probabilities are projected one step ahead and translated into portfolio weights through regime-conditional mean–variance optimization. Third, following the caution in Ilmanen et al. (2021) from AQR, VIX and CAPE are not used as standalone timing signals or as inputs to the HMM. Instead, they enter only through the optimizer’s reference allocation w_0 , where they act as regime-conditioned stabilization anchors. Finally, the entire pipeline is evaluated under a strict burn-in and walk-forward protocol, so that configuration selection, regime inference, anchor estimation, and portfolio optimization are separated from the held-out testing period.

Viewed this way, the main finding is not simply that latent regimes exist, nor that macro variables can time factor premia directly. Rather, the evidence suggests that regime information becomes economically useful when it is connected to implementable portfolio construction, and that macro-conditioning variables can add value when used to stabilize optimization rather than to forecast returns directly.

7.1 Regime Asymmetry and Factor Usage

A central empirical pattern is that the regimes are not simple mirror images of one another. The model does not merely separate the sample into “good” and “bad” markets. Instead, different regimes are associated with different factor payoffs, and some factors appear useful only under specific states or only through short-side exposure.

The factor exposure analysis supports this interpretation. Both final strategies share a common core exposure to market, quality, low-risk, and low-leverage factors. This common core explains why both strategies maintain stable long-run performance. However, the two specifications use the remaining factors very differently. The long-only strategy concentrates in a smaller set of consistently attractive factors and largely avoids factors such as accruals, seasonality, short-term reversal, and size. Under the no-shorting constraint, the optimizer expresses negative views by assigning little or no weight to weaker factors.

By contrast, the long-short aggressive strategy uses nearly all factors more actively, including negative positions in factors that are valuable mainly on the short side in certain regimes.

This suggests that regime-aware allocation is not only about increasing or decreasing total risk exposure; it is also about changing the cross-sectional role of individual factors. Some factors may be useful as long exposures in one regime, irrelevant in another, and useful as short exposures in a third. This asymmetry is precisely why a regime-conditional allocation framework can add value beyond a static multifactor portfolio.

This also explains the turnover difference between the two strategies. Because the aggressive strategy can both enter and exit short positions across regime transitions, it exhibits more dynamic allocation changes and substantially higher turnover. The long-only strategy, in contrast, delivers a cleaner defensive smart-beta profile: lower volatility, lower concentration, lower turnover, and stronger drawdown control.

7.2 The Role of the Anchor

The anchor mechanism should not be interpreted as a universal risk-reduction device. Its contribution depends on the investor mandate and the feasible weight set. In the long-only specification, the CAPE hard-regression anchor improves the Sharpe ratio from 1.16 to 1.33 while leaving annualized excess return nearly unchanged. The improvement mainly comes from risk control: volatility declines from 4.61% to 4.05%, maximum drawdown improves from -6.43% to -3.64% , and the Calmar ratio increases from 0.83 to 1.48. Thus, for the long-only strategy, the anchor acts primarily as a stabilizer.

For the aggressive long-short specification, the VIX soft-regression anchor plays a different role. It increases annualized excess return from 6.15% to 6.86% and slightly improves the Sharpe ratio from 1.22 to 1.24, but it also increases volatility, drawdown, and turnover. This is consistent with the strategy’s objective: the aggressive specification is designed to seek higher return by allowing short positions and more flexible allocation. In this case, the anchor improves the return-seeking profile rather than uniformly reducing risk.

This distinction is important for interpreting the paper’s contribution. The anchor is not a direct macro timing signal in the AQR-style sense. We do not ask whether VIX or CAPE can independently forecast factor premia. Instead, the anchor helps the optimizer choose more stable and economically meaningful weights conditional on the HMM-implied regime structure. In other words, the macro variable is useful not because it directly predicts returns, but because it helps align portfolio optimization with the market environment inferred from the factor panel.

7.3 Performance During Stress and Transition Periods

The benchmark comparison shows that the proposed strategies do not maximize raw return during strong equity bull markets. The Nasdaq and S&P 500 achieve higher cumulative growth over the post-2013 expansion, particularly in 2019, 2021, and 2023. The regime-aware strategies, however, produce a substantially smoother return path with smaller drawdowns. This suggests that the framework’s primary advantage lies in risk-adjusted performance rather than full participation in high-beta equity rallies.

The strongest evidence in favor of the framework appears during stress periods. In 2020, the long-short aggressive strategy outperforms the S&P 500 by 6.27%, while the long-only strategy also remains resilient. In 2022, during the joint equity–bond drawdown, the aggressive and long-only strategies generate returns of 12.47% and 6.98%, respectively, while SPY (-18.26%), Nasdaq (-32.65%), and long-duration Treasury bonds (-31.40%) decline sharply. These episodes directly support the central motivation of the paper: static diversification can fail when correlations rise and regimes shift, whereas regime-aware factor allocation adapts exposures more effectively to the prevailing state.

At the same time, the results reveal an important limitation. The framework performs less strongly during “Walking on Ice” periods, which tend to occur immediately before or after major crisis states. These periods are characterized by sharp rebounds, factor reversals, and

unstable transitions between defensive and high-beta behavior. Because the HMM identifies these observations as closely related to crisis-adjacent conditions, the portfolio remains cautious and fails to fully capture post-crisis upside. This is not a failure of the regime model so much as a manifestation of an inherent trade-off: a framework designed to avoid crisis losses will naturally underparticipate in abrupt recoveries.

This pattern also explains why pre-crisis and post-crisis episodes are assigned to related regimes. Both environments share elevated uncertainty, unstable factor leadership, and rapid changes in market direction, so the similarity is economically meaningful even though it costs the strategy some recovery gains. This is consistent with Botte and Bao (2021), who find that clustering-based regime models also group pre- and post-crisis observations into a single state, accepting missed recovery participation in exchange for stronger downside protection. A natural extension, therefore, is to introduce an additional conditioning variable that can distinguish “walking-on-ice before crisis” from “walking-on-ice after crisis” regimes. Doing so would preserve the framework’s downside protection while allowing it to re-engage risk more quickly during recoveries.

7.4 Practical Interpretation

From an implementation perspective, the two final specifications serve different roles. The long-only strategy is the more natural core allocation. It delivers the strongest Sharpe ratio, lowest volatility, lowest drawdown, and lower turnover. Its performance profile is consistent with a defensive smart-beta allocation that seeks stable excess return rather than aggressive capital growth.

The long-short strategy is better interpreted as a return-seeking satellite allocation. It produces a higher annualized total return, but this comes with higher volatility, deeper drawdowns, higher turnover, and greater implementation complexity. The higher turnover also makes the aggressive strategy more sensitive to transaction costs, shorting costs, financing costs, and the availability of liquid instruments for factor replication.

Therefore, the two strategies should not be ranked by return alone. The long-only strategy is more attractive for investors prioritizing stability, drawdown control, and implementability. The aggressive long-short strategy is more attractive for investors willing to accept higher trading intensity and larger interim losses in exchange for higher return potential.

7.5 Limitations and Future Extensions

Several limitations remain. First, the model is estimated using monthly data, which is appropriate for a smart-beta allocation framework but limits the number of observations available for fitting regime-specific parameters. This motivates the use of diagonal HMM covariance matrices and Ledoit–Wolf shrinkage, but also restricts the ability to estimate richer covariance structures or conduct a fully strict expanding-window validation within the burn-in period.

Second, the framework assumes Gaussian emission distributions. Although the mixture structure can partially absorb fat tails through high-variance regimes, factor returns are empirically non-Gaussian. Future work could examine Student- t HMMs or other heavy-tailed state-space models to better capture extreme factor-return realizations.

Third, implementation frictions could be modeled more precisely. This paper uses a first-order transaction-cost estimate based on weighted basis-point assumptions for ETF-sourced and equity-replicated factor sleeves. A more rigorous extension could follow Frazzini, Israel, and Moskowitz (2018) by assigning factor-specific costs based on liquidity, turnover, bid–ask spreads, and market impact. Future work could compute realized turnover directly from the rebalancing path,

$$\text{Turnover}_t = \sum_i |w_{i,t} - \tilde{w}_{i,t}|,$$

where $\tilde{w}_{i,t}$ is the pre-rebalancing weight after return-driven portfolio drift, and deduct costs from monthly returns before recomputing net performance.

Fourth, the robustness analysis can be further expanded. The current checks validate the effectiveness of the HMM relative to the GMM and the importance of the VIX/CAPE anchors. A useful extension would add a static mean–variance benchmark that uses the same factor universe and optimization objective but holds weights fixed out of sample, thereby isolating the incremental value of regime-dependent allocation. Future versions could also include bootstrap or standard-error-based significance tests for Sharpe ratios, returns, drawdowns, and other performance statistics.

Overall, the results suggest that regime-aware multifactor allocation is most useful when treated as a disciplined portfolio-construction problem rather than as a pure return-forecasting exercise. The HMM provides probabilistic regime information, the optimizer translates this information into factor weights, and the anchor stabilizes allocations without allowing macro variables to dominate regime identification. The resulting framework improves the stability and resilience of factor allocation across changing market environments.

8 Conclusion

This paper develops a regime-conditioned downside-aware multifactor allocation framework that connects latent market-state inference to implementable portfolio construction. Motivated by the limited out-of-sample performance of traditional factor-timing signals and by prior evidence that latent regimes can be recovered from factor-return panels, the framework extends static GMM-style regime classification into a Gaussian Hidden Markov Model. This allows regime persistence and transition dynamics to be estimated directly, so that portfolio weights are formed from one-step-ahead regime probabilities rather than contemporaneous regime labels.

The main contribution is not simply identifying market regimes, but converting regime information into portfolio decisions. The HMM provides a probabilistic description of the market environment, while the regime-conditional optimizer translates that information into factor weights. VIX and CAPE are used as auxiliary optimization anchors: they stabilize the optimizer’s reference allocation without entering the regime-identification step or serving as standalone return predictors. This design separates regime inference from macro timing and treats factor premia as state-dependent allocation opportunities rather than fixed static exposures.

The empirical results support this interpretation. Over the 2013–2023 walk-forward evaluation period, the long-only protective strategy achieves a Sharpe ratio of 1.33, annualized excess return of 5.39%, and maximum drawdown of -3.64% . The long-short aggressive strategy produces higher annualized excess return of 6.86%, with a Sharpe ratio of 1.24 and maximum drawdown of -5.78% . While passive equity benchmarks earn higher raw returns during strong bull-market years, the regime-aware strategies deliver smoother return paths, shallower drawdowns, and stronger downside control. Their resilience during the 2020 COVID shock and the 2022 joint equity–bond drawdown suggests that the framework’s value lies primarily in risk management and regime-conditioned factor allocation rather than raw-return maximization.

The two final specifications serve different practical roles. The long-only strategy is better suited as a core defensive smart-beta allocation because it has lower volatility, lower drawdown, lower turnover, and stronger net-of-cost robustness. The long-short strategy is better interpreted as a return-seeking satellite allocation: it earns higher gross return but requires more active trading, allows short positions, and is more sensitive to transaction costs, financing costs, and implementation frictions. The preferred specification therefore depends on the investor’s objective rather than on raw return alone.

Several limitations remain. The monthly frequency is appropriate for realistic smart-beta implementation but limits the effective sample size for regime-specific estimation. The Gaussian emission assumption is tractable and interpretable, but future work could examine Student- t HMMs or other heavy-tailed state-space models. Transaction costs are also treated through a first-order basis-point approximation; a more rigorous implementation could compute realized turnover directly from the rebalancing path, assign factor-specific trading costs, and deduct those costs before recomputing net performance. Future work could also add statistical validation for Sharpe ratios, drawdowns, and performance differences across dynamic and static benchmarks.

Overall, the results suggest that regime-aware multifactor allocation is most effective when framed as a portfolio-construction problem rather than a pure return-forecasting exercise. Factor premia need not be abandoned simply because standalone timing signals are fragile; instead, they can remain economically useful when regime information is translated into disciplined, turnover-aware, and risk-controlled allocation decisions.

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Appendix

A.1 Implementation and Reproducibility Notes

This appendix summarizes the main implementation details used to reproduce the empirical results. All random procedures use `random_state = 61`. The same seed is used for GMM initialization, HMM fitting, and all repeated model-selection procedures whenever applicable.

The HMM is warm-started from the fitted GMM parameters. In the initial HMM fit, `init_params=""` is used to suppress `hmmlearn`'s default re-initialization and preserve the GMM-implied starting probabilities, means, and covariance estimates. In the walk-forward refitting stage, `init_params="c"` is used so that the covariance matrices are re-estimated from the expanding data at each step while the initialized starting probabilities, means, and transition matrix are preserved.

All sample covariance matrices are symmetrized before optimization:

$$\hat{\Sigma} \leftarrow \frac{1}{2} \left(\hat{\Sigma} + \hat{\Sigma}^\top \right).$$

A small diagonal jitter of $10^{-6}I$ is added when necessary to avoid numerical positive-semidefinite failures in CVXPY and to improve solver stability. Ledoit–Wolf shrinkage is used in the final reported strategies because it provides a more stable covariance estimate than the raw sample covariance in the monthly factor panel.

The portfolio optimization is solved using SLSQP in the final reported configurations. CVXPY-based solvers, including OSQP, CLARABEL, and ECOS, were also considered during the burn-in grid search. When a candidate optimization fails to converge or produces infeasible weights, that candidate is excluded from the configuration ranking.

A.2 Factor Universe

Table A1 reports the factor universe used in the empirical analysis. The factors are selected to cover a broad set of equity-premia themes, including market exposure, valuation, size, momentum, profitability, quality, risk, investment, financing activity, reversal, and seasonality. Returns are used in raw form for portfolio construction and standardized only for GMM/HMM regime estimation.

Table A1: Factor universe used in the regime-aware allocation framework.

Factor	Theme	Interpretation
<code>accruals</code>	Accounting quality	Firms with lower accruals relative to earnings
<code>debt_issuance</code>	Financing activity	Firms with lower net debt issuance
<code>investment</code>	Investment	Firms with lower asset growth or investment intensity
<code>low_leverage</code>	Balance-sheet risk	Firms with lower leverage
<code>low_risk</code>	Defensive / risk	Lower-beta or lower-volatility firms
<code>mkt</code>	Market	Broad equity market excess return
<code>momentum</code>	Trend	Recent winners relative to recent losers
<code>profit_growth</code>	Growth quality	Firms with stronger profitability growth
<code>profitability</code>	Profitability	More profitable firms
<code>quality</code>	Quality	Firms with stronger quality characteristics
<code>seasonality</code>	Seasonal pattern	Firms with persistent seasonal return effects
<code>short_term_reversal</code>	Reversal	Recent losers relative to recent winners over short horizons
<code>size</code>	Size	Small-cap firms relative to large-cap firms
<code>value</code>	Valuation	Cheap firms relative to expensive firms

A.3 Hyperparameter Search Space

Table A2 summarizes the main hyperparameter grid considered during the burn-in selection stage. All final choices are selected using only the burn-in period and then held fixed during the 2013–2023 walk-forward evaluation.

Table A2: Main hyperparameter search space.

Component	Candidate values
Number of regimes K	$\{2, 3, 4, 5, 6, 7\}$
HMM covariance type	Full, diagonal
Transition diagonal strength d	$\{0.20, 0.22, 0.23, 0.24, 0.25, 0.30, 0.40\}$
Covariance estimator	Sample, Ledoit–Wolf, exponentially weighted
Risk aversion λ	$\{0.2, 0.3, 0.4, 0.5\}$
Turnover/anchor penalty α	0.01
Minimum weight $w_{\min}$	$\{-0.2, -0.1, -0.05, 0\}$
Maximum weight $w_{\max}$	$\{0.4, 0.5\}$
Solver	OSQP, CLARABEL, ECOS, SLSQP
Anchor variable	None, VIX, CAPE
Anchor type	Non-regression, regression-based
Anchor mode	Global, hard, soft

The final reported long-only strategy uses $K = 4$, diagonal HMM covariance, Ledoit–Wolf covariance shrinkage, SLSQP optimization, $w_{\min} = 0$, $w_{\max} = 0.5$, and a CAPE hard-regression anchor. The final reported long-short strategy uses the same HMM specification but allows $w_{\min} = -0.2$ and uses a VIX soft-regression anchor.

A.4 Portfolio Objective and Constraints

For each regime r , the regime-conditional portfolio weight is obtained from the following turnover-aware mean–variance problem:

$$\min_{w \in \mathbb{R}^N} -\mu_r^\top w + \lambda w^\top \Sigma_r w + \alpha \|w - w_0\|_1$$

subject to

$$\mathbf{1}^\top w = 1, \quad w_{\min} \leq w_i \leq w_{\max}, \quad i = 1, \dots, N.$$

Here μ_r and Σ_r denote the regime-specific expected return vector and covariance matrix. The term $\lambda w^\top \Sigma_r w$ penalizes portfolio risk, while $\alpha \|w - w_0\|_1$ penalizes deviations from the reference allocation. During burn-in, w_0 acts primarily as a regularization reference. During walk-forward deployment, w_0 is replaced by the VIX- or CAPE-conditioned anchor, so the same term functions as a turnover-aware stabilization mechanism.

The deployed portfolio weight is formed using the one-step-ahead HMM predictive regime distribution:

$$w_{t+1} = \sum_{r=0}^{K-1} \hat{\pi}_{t+1|t}(r) w_r, \quad \hat{\pi}_{t+1|t} = \gamma_t^\top T,$$

where γ_t is the filtered regime-probability vector and T is the estimated transition matrix.

A.5 Performance Metric Definitions

Let $r_{p,t}$ denote monthly portfolio excess return and T denote the number of monthly observations in the evaluation period. Annualized excess return is computed as

$$R_{\text{ann}} = \left(\prod_{t=1}^T (1 + r_{p,t}) \right)^{12/T} - 1.$$

Annualized volatility is computed as

$$\sigma_{\text{ann}} = \sqrt{12} \cdot \text{Std}(r_{p,t}).$$

The Sharpe ratio is computed using excess returns:

$$\text{Sharpe} = \frac{R_{\text{ann}}}{\sigma_{\text{ann}}}.$$

Maximum drawdown is computed from the cumulative wealth process

$$W_t = \prod_{\tau=1}^t (1 + r_{p,\tau}),$$

as

$$\text{MDD} = \min_t \left(\frac{W_t}{\max_{\tau \leq t} W_\tau} - 1 \right).$$

The Calmar ratio is computed as

$$\text{Calmar} = \frac{R_{\text{ann}}}{|\text{MDD}|}.$$

The monthly win rate is the fraction of months with positive portfolio return:

$$\text{WinRate} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{r_{p,t} > 0\}.$$

A.6 Turnover Annualization

Turnover is measured using one-way portfolio turnover. For a rebalance date t , the one-way turnover is

$$\text{TO}_t = \frac{1}{2} \sum_{i=1}^N |w_{i,t} - w_{i,t-1}|.$$

Because the final strategies rebalance every two months, there are approximately six rebalance dates per year. Annualized turnover is therefore computed as

$$\text{TO}_{\text{ann}} = 6 \cdot \mathbb{E}[\text{TO}_t].$$

Equivalently, if the reported turnover per rebalance is computed using the full absolute weight change $\sum_i |w_{i,t} - w_{i,t-1}|$, then annualized one-way turnover is

$$\text{TO}_{\text{ann}} = 3 \cdot \mathbb{E} \left[\sum_{i=1}^N |w_{i,t} - w_{i,t-1}| \right],$$

where the factor $3 = 6/2$ combines the six bi-monthly rebalances per year with the two-way-to-one-way conversion.

A.7 Transaction-Cost Approximation

The transaction-cost analysis uses a first-order weighted basis-point approximation. Let TO_{ETF} denote annual turnover attributed to ETF-source exposures and $\text{TO}_{\text{Equity}}$ denote annual turnover attributed to equity-replicated exposures. The annual cost drag is approximated as

$$\text{Cost} = c_{\text{ETF}} \cdot \text{TO}_{\text{ETF}} + c_{\text{Equity}} \cdot \text{TO}_{\text{Equity}},$$

where the base-case assumptions are

$$c_{\text{ETF}} = 5 \text{ bps}, \quad c_{\text{Equity}} = 30 \text{ bps}.$$

Net annualized excess return is then approximated by

$$R_{\text{net}} = R_{\text{gross}} - \text{Cost}.$$

This approximation is intentionally conservative and transparent. It does not fully model borrow fees, short rebates, financing costs, market impact, or factor-specific liquidity differences. A more detailed implementation would compute realized turnover directly from the monthly rebalancing path, assign factor-specific costs, deduct trading costs from each monthly return, and then recompute net annualized return, volatility, Sharpe ratio, and drawdown.

A.8 Notes on Static and Dynamic Benchmarks

The static equal-weight benchmark holds a constant $1/N$ allocation across the same fourteen-factor universe:

$$w_i = \frac{1}{N}, \quad i = 1, \dots, N.$$

This benchmark contains no regime inference, no covariance estimation, no macro anchor, and no dynamic optimization. It is therefore useful for isolating whether the regime-aware framework adds value beyond simple diversification across the selected factor universe.

The HMM and GMM robustness comparison isolates the role of transition dynamics. The GMM can recover latent clusters, but it does not estimate regime persistence or provide a transition-based one-step-ahead regime forecast. The HMM is therefore more consistent with the walk-forward allocation rule because it maps filtered probabilities into predictive probabilities through $\gamma_t^\top T$.